

Avian-specific *Salmonella enterica* Serovar Gallinarum transition to endemicity is accompanied by localized resistome and mobilome interaction

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This study uses a large dataset from both recent isolates and genomes in databases to provide an **important** analysis of the population structure of the pathogen *Salmonella gallinarum*. The authors present **convincing** results regarding the regional adaptation and the evolutionary trajectory of the resistome and mobilome, even though some issues regarding the genomic analysis could be improved. This work will interest microbiologists and researchers working on genomics, evolution, and antimicrobial resistance.

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Abstract

Bacterial regional demonstration after global dissemination is an essential pathway for selecting distinct fitnesses. However, the evolution of the resistome during the transition to endemicity remains unaddressed. Using the most comprehensive whole-genome sequencing dataset of *Salmonella enterica* serovar Gallinarum (*S. Gallinarum*) collected from 15 countries, including 45 newly recovered samples from two related local regions, we established the relationship among avian-specific pathogen genetic profiles and localization patterns. Initially, we revealed the international transmission and evolutionary history of *S. Gallinarum* to recent endemicity through phylogenetic analysis conducted using a spatiotemporal Bayesian framework. Our findings indicate that the independent acquisition of the resistome via the mobilome, primarily through plasmids, transposons, and prophages,

shapes a unique antimicrobial resistance profile among different lineages. Notably, the mobilome-resistome combination among distinct lineages exhibits a geographical-specific manner, further supporting a localized endemic mobilome-driven process. Collectively, this study elucidates resistome adaptation in the endemic transition of an avian-specific pathogen, likely driven by the localized farming style, and provides valuable insights for targeted interventions.

Introduction

The acquisition of antimicrobial resistance (AMR) by pathogens is well-established as one of the most severe threats of the 21st century (1–3). It is estimated that over 10 million people could die per year due to AMR by 2050 (4). Antimicrobial resistance genes (ARGs), collectively known as the resistome, play a pivotal role in AMR development, progression, and amplification (5). Pathogens typically acquire resistome through mobilomes, like plasmids, via horizontal gene transfer (HGT) (6, 7). From an evolutionary perspective, the dynamic change in the resistome is crucial for enhancing pathogen fitness and enabling it to colonize new ecological niches (8). Key factors contributing to resistome alterations include disease burden, human activities, climate change, and geographical selection forces (9–13). Understanding the evolutionary trajectory of the AMR and related reservoirs is necessary, as disease management strategies will differ substantially (14). However, knowledge regarding the overall profile of pathogen regional adaptation related to the stepwise dynamics of the resistome remains limited.

Before recent advancements in whole-genome sequencing (WGS) technology, little was understood about resistome diversity that affects pathogen endemicity (15). Traditional typing methods generally exhibit low resolution, making monitoring and quantitatively comparing horizontal ARG transfer events challenging. The scarcity of appropriate models also poses a limitation. *Salmonella*, a pathogen with distinct geographical characteristics, has more than 90% of its serovars frequently categorized as geo-serotypes (16). Among the thousands of geo-serotypes, *Salmonella enterica* serovar Gallinarum (*S. Gallinarum*) is an avian-specific pathogen that causes severe mortality, with particularly detrimental effects on the poultry industry in low- and middle-income countries (17, 18). As once a globally prevalent pathogen in the 20th century, *S. Gallinarum* was listed by the World Organization for Animal Health (WOAH) and gradually became an endemic pathogen with sporadic outbreaks following the implementation of eradication programs in most high-income countries, making it a perfect model for study (19, 20).

Nowadays, antimicrobial therapy remains a priority choice against *S. Gallinarum* infections (21). The overuse or misuse of antimicrobials has led to increased epidemiological escalation of *S. Gallinarum*, with a regionally higher risk of AMR, especially for sulfonamides, penicillin, and tetracyclines (22, 23). To fill the gaps in understanding the evolution of *S. Gallinarum* under regional-associated AMR pressures and its adaptation to endemicity, we collected the most comprehensive set *S. Gallinarum* isolates, consisting of 580 genomes, spanning the period from 1920 to 2023. Using such unique WGS dataset, we investigated: 1) the population structure and potential geographical transmission history of *S. Gallinarum* at a single base level; 2) the dynamic resistome and mobilome changes that are associated with *S. Gallinarum* transition to an endemic variant; and 3) horizontal resistome transfer frequency and pattern between distinct regions.

Results

Global distribution of *S. Gallinarum* links with distinct sublineages

To understand the global geographic distribution and genetic relationships of *S. Gallinarum*, we assembled the most comprehensive *S. Gallinarum* WGS dataset ($n=580$), comprising 535 publicly available genomes and 45 newly sequenced genomes (24 [Fig.](#)). The core-genome single nucleotide polymorphism sites (cgSNPs) were obtained by using the fully sequenced genome of *S. Gallinarum* R51 as a reference. Through hierarchical Bayesian analysis of cgSNPs, it was confirmed that *S. Gallinarum* divides into three biovars: *S. Gallinarum* biovar Pullorum (bvSP) ($n=528/580$, 91.03%), *S. Gallinarum* biovar Gallinarum (bvSG) ($n=50/580$, 8.6%), and *S. Gallinarum* biovar Duisburg (bvSD) ($n=2/580$, 0.34%). An association was identified between the biovar type of *S. Gallinarum* and its global geographic distribution. Most bvSP isolates were from Asia (436/528), with fewer occurrences in Europe (54/528), South America (12/528), and North America (5/528). In bvSG, South America (18/50, 36%) is the primary source of isolates, followed by North America (9/50, 18%), Europe (9/50, 18%), Africa (7/50, 14%), and Asia (3/50, 6%) (S1 Fig).

From a lineage perspective, bvSP can be further classified into five lineages: L1 (10/528), L2a (45/528), L2b (163/528), L3b (169/528), and L3c (141/528). Importantly, L3a, previously considered distinct, has been revealed as a sub-lineage of L3c. The predominant lineage types vary across different continents. Regarding predominant bvSP sources, Asia stands out, with L3c, L3b, and L2b being the main lineage types. On the other hand, L3b and L2a are more widespread in Europe and America, respectively (Fig 1a [Fig.](#)). For the bvSP strains from Asia included in our dataset, we found that all originated from China. To further investigate the distribution of bvSP across different regions in China, we categorized them into three distinct regions: southern, eastern, and northern (Supplementary Table 3). Our findings indicate significant variations in bvSP across various regions from geo-temporal aspects. Prevalence was highest in the eastern region (233/436), with lower incidences in the northern (137/436) and southern regions (65/436). Interestingly, the predominant bvSP lineage type also varied. In eastern and southern China, L2b and L3c exhibited as the predominant lineage types, while in northern China, L3b and L3c held nearly equivalent positions (Fig 1c [Fig.](#)).

An analysis of the temporal prevalence of bvSP in China revealed a gradual replacement pattern of the lineages. Before the 2000s, the predominant lineage type was L3b. Then, from the 2000s to the 2010s, L3b and L1 were continually replaced by L3c, with L2b becoming the predominant lineage after the 2010s (Fig 1b [Fig.](#)). However, the replacement pattern varied among different regions. Specifically, the same pattern was observed in eastern China, but for northern and southern China, only replacements from L3b to L2b and L1 to L3c were observed (S2 Fig). These findings might indicate accelerated localized adaptation of bvSP.

Genomics portrait bvSP historic transmission

Considering the historical pandemic status of bvSP, we then investigated its global geographic transmission routes to understand its evolution into an endemic pathogen. L2b and L3b were identified as the dominant global lineages due to their antimicrobial resistance risks, and potential intercontinental transmission events. Other lineages were also analyzed using a Bayesian model with a relaxed molecular clock to infer their historical evolution (S3-S5 Figs). The temporal structure was verified (S6 Fig). Our findings show that the origin of L3b in China can be traced back to as early as 1683 (95% CI: 1608 to 1839). In contrast, the earliest possible origin of L2b in China dates back to 1880 (95% CI: 1838 to 1902) (Fig 2 [Fig.](#)). Then, we specifically estimated the time points for the first intercontinental transmission events for the two major lineages. Our results indicate that L2b, likely underwent two major intercontinental transmission events. The first occurred around 1893 (95% CI: 1870 to 1918), with transmission from China to South America. The second major transmission event occurred in 1923 (95% CI: 1907 to 1940), involving the spread from South America to Europe. In contrast, the transmission pattern of L3b, an *S. Gallinarum*

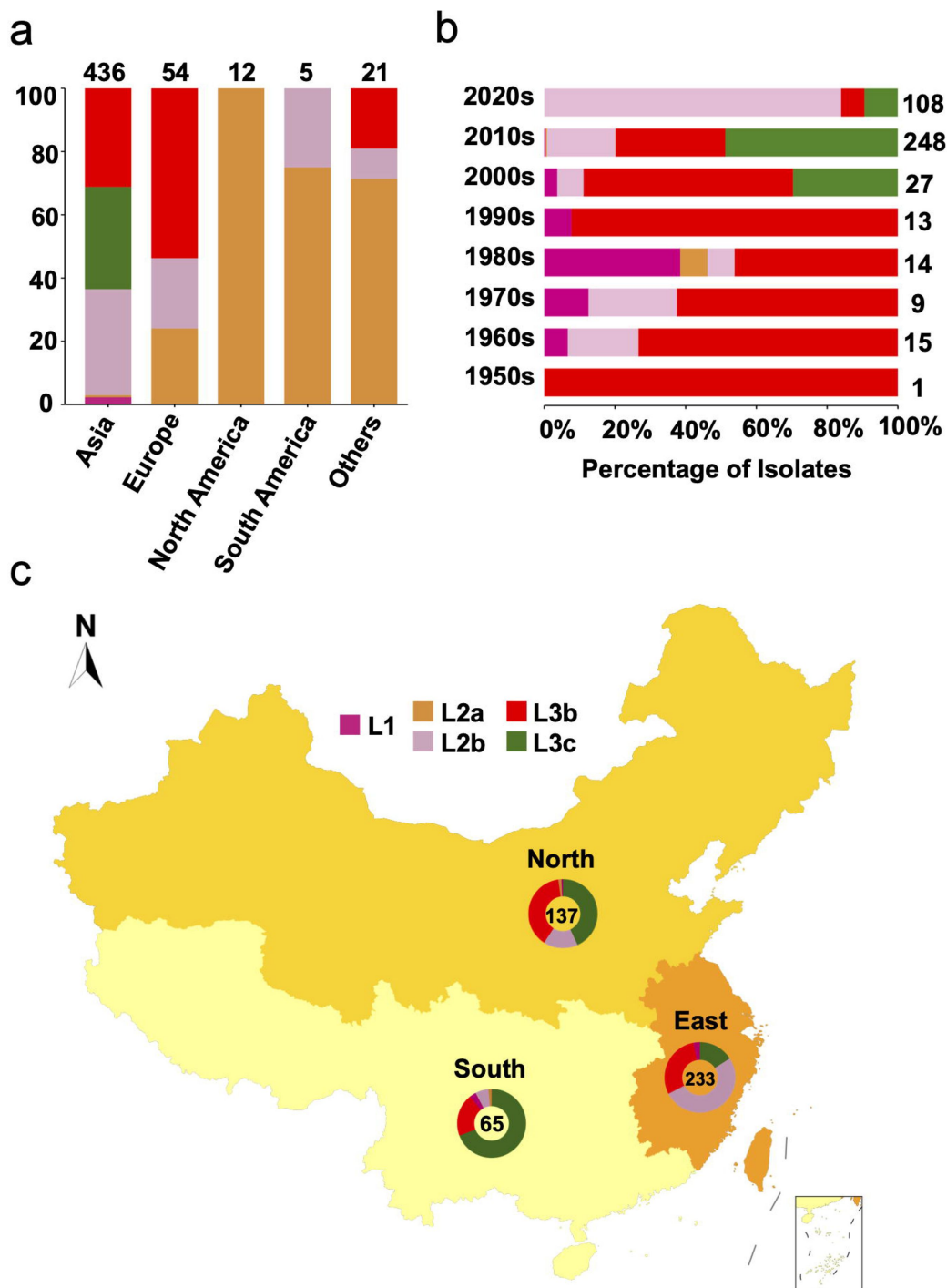


Figure 1.

Genetic diversity of *S. Gallinarum* biovar Pullorum (bvSP) by geography and time.

Different colours were used to represent the various lineages of bvSP: fuchsia for L1, orange for L2a, pink for L2b, red for L3b, and green for L3c. (a) The bvSP in the dataset is classified into five continents based on their isolation regions. The bar graph illustrates the distribution of lineage-specific bvSP across continents, depicted as percentages. The total sample size for each bar is indicated at the top. (b) The bar graph shows the distribution of bvSP isolated in China by lineage per decade. The total sample size for each bar is also indicated on the right side. (c) Geographic distribution of bvSP isolated from China. Doughnut charts in the map show the proportion of lineage types of bvSP collected in the corresponding region, with the total number of isolates in the center.

lineage originating in China, appears relatively more straightforward. It underwent only one intercontinental transmission event, from China to Europe, likely around 1790 (95% CI: 1661 to 1890) (S7 Fig).

Endemic isolates further support localized dissemination by bvSP

To investigate the dissemination pattern of bvSP, we obtained forty-five newly isolated bvSP from 734 samples (6.1% overall isolation rate) collected from diseased chickens at two farms in Yueqing and Taishun, Zhejiang Province. Those bvSP can be classified into two dominant sequence types (STs): ST3717 and ST92. Isolates from Yueqing were all identified as ST3717, while most Taishun isolates were characterized as ST92. Only the E404 isolate was identified as ST2151 (**Fig 3a** [↗](#)). Of note, a lineage-preferential association was also observed, with bvSP isolated from Taishun belonging to L3b and those from Yueqing belonging to L2b. To further understand the genetic relationship among isolates in the two regions, we calculated cgSNPs distances between isolates. Isolates with cgSNP distances less than five were determined as the threshold to be involved in the same transmission event. Interestingly, genetic clustering revealed that isolates from the same farm exhibited a significantly close genetic correlation (**Fig 3b** [↗](#)). No historical transmission events of bvSP were found between Taishun and Yueqing.

Other genetic characteristics also exhibited a similar pattern. The ARG and plasmid profile disclosed that *IncFII(S)* and *ColpVC* plasmids exhibited the highest carriage percentages (100%) among the 45 bvSP isolates. The *IncX1* plasmid was exclusively identified in the E404 isolate. Furthermore, E404 harboured two distinctive resistance genes, *bla*_{TEM-1B} and *sul2*, in contrast to the absence of ARGs in the remaining bvSP isolates. We also estimated the invasiveness index to assess the invasiveness of isolated bvSP and showed that bvSP from the same farm had comparable invasive abilities. However, isolates from Taishun demonstrated a higher invasive ability than those from Yueqing (**Fig 3c** [↗](#), Supplementary Table 6).

Furthermore, we simulated potential transmission events between the bvSP strains isolated from Zhejiang Province (n=95) and those from China with available provincial information (n=435) using five SNPs as the threshold. As a result, the transmission events showed a strong geographic-preferential distribution. We identified a total of 91 potential transmission events, all of which occurred exclusively within Zhejiang Province. No inter-provincial transmission events were detected, supporting the statement that bvSP in China is a highly localized pathogen (S8 Fig, Supplementary Table 8).

Mobilome and resistome distinct lineage-preferential association in a regional manner

The mobilome drives partitions of the resistome and plays a pivotal role in shaping the ecological niches and local adaptation of bacteria. Therefore, a quantitative evaluation of the resistome and mobilome is necessary to enhance the understanding of the localized distribution of bvSP. For *S. Gallinarum*, a total of 13 classes of ARGs, which can be further classified into six categories, were identified. The results revealed that bvSP exhibited a significantly greater resistome than bvSG and bvSD (**Fig 4a** [↗](#)). Among them, *sul2* (196/528, 37.1%) has the highest prevalence, followed by *bla*_{TEM-1B} (183/528, 34.7%) and *tet(A)* (104/528, 19.7%). Moreover, the resistome also demonstrated a lineage-preferential association. We observed that L3b is more inclined to carry *bla*_{TEM-1B} and *sul2* than other lineages, whereas *tet(A)* exclusively exists within L3 (Supplementary Table 4). The diversity of resistome carried by L3 may elevate the risk of multi-drug resistance (MDR). Notably, AMR risks vary among different regions of China, with the highest risk observed in southern China, followed by the northern and eastern regions (**Fig 4c** [↗](#)). Interestingly, *tet(A)*, *bla*_{TEM-1B}, and *sul2* were predominant resistome types prevalent across all regions of China. Meanwhile, *aadA5* was typically more commonly found in southern China (**Fig 4b** [↗](#)).

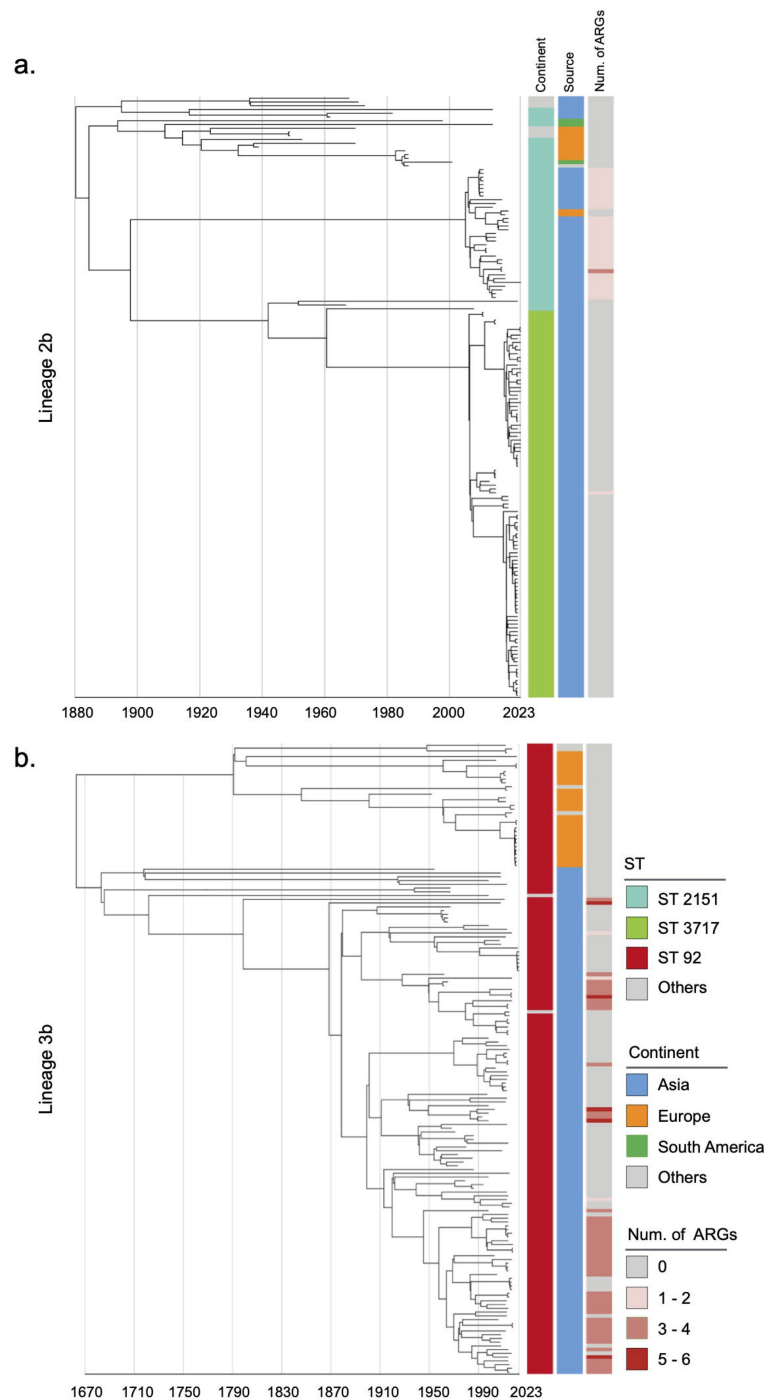


Figure 2.

Phylogenetic tree of *S. Gallinarum* L2b and L3b based on a spatiotemporal Bayesian framework.

The phylogenetic tree on the left was constructed using a reference-mapped multiple core-genome SNPs sequence alignment, with recombination regions detected and removed by Gubbins. The spatiotemporal Bayesian framework was configured with the "GTR" substitution model, 4 Gamma Category Count, "Relaxed Clock Log Normal" model, "Coalescent Bayesian Skyline" tree prior model, and a Markov Chain Monte Carlo (MCMC) chain length of 100,000,000, with sampling every 10,000 iterations. Convergence was assessed using Tracer, ensuring all parameter effective sampling sizes (ESS) exceeded 200. Evolutionary time is represented by the length of the branches. The heatmap on the right displays, respectively, the sequence type (ST), region of isolation, and the number (Num.) of antimicrobial resistance genes (ARGs) carried by the corresponding *Salmonella* Gallinarum.

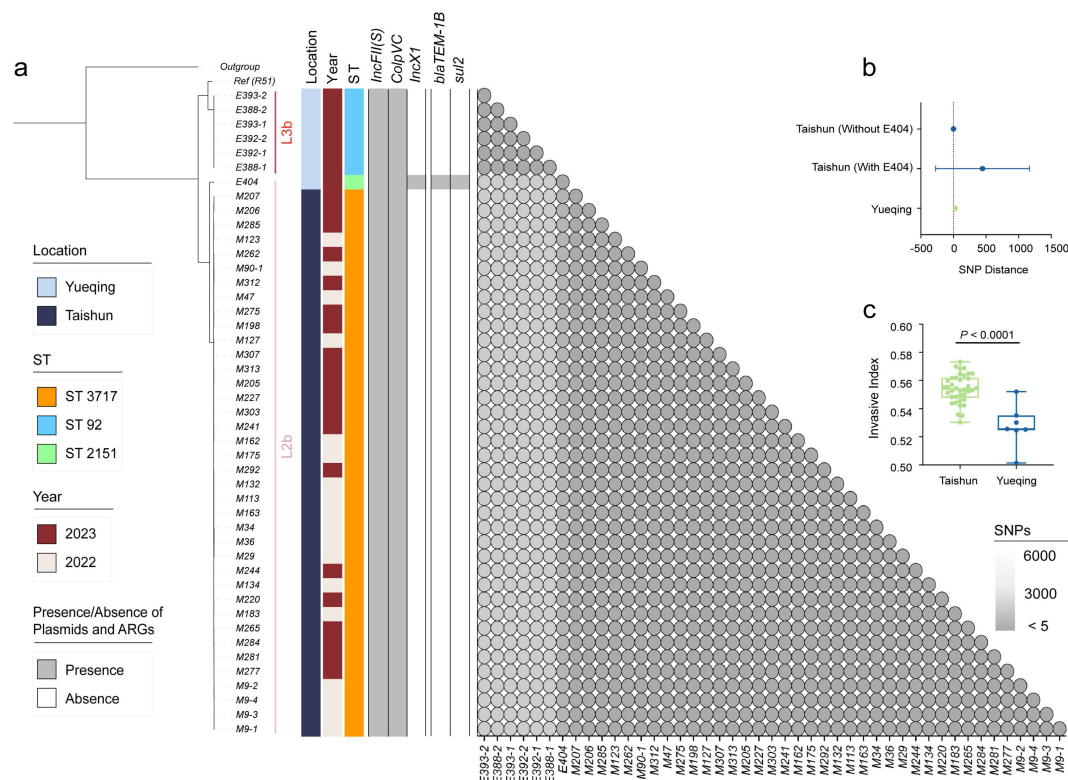


Figure 3.

Genomic characteristics of 45 newly isolated *S. Gallinarum* biovar Pullorum (bvSP) strains from Taishun and Yueqing, Zhejiang Province, China.

(a) The phylogenetic tree, constructed using cgSNPs, categorizes the 45 bvSP strains into two distinct lineages (L3b and L2b). On the left side of the heatmap, information on isolation regions, sequence types (ST), and sampling times are displayed, with various colors indicating different categories as specified on the left. The right section of the heatmap presents a detailed matrix showing plasmids, ARGs, and cgSNP distances. The presence of a plasmid or ARG in an isolate is denoted by gray shading, while absence is indicated by white. cgSNPs below five were used as the threshold, with darker circles signifying a higher probability of transmission between isolate pairs. (b) The average cgSNP distance between isolates from Taishun and Yueqing. E404 led to an increase in the mean cgSNP distance of bvSP from Taishun. (c) Invasiveness index of bvSP in Taishun and Yueqing. The results show a higher invasiveness index for bvSP isolated from Taishun, indicating that bvSP isolated from Taishun might have greater invasive capabilities among vulnerable hens.

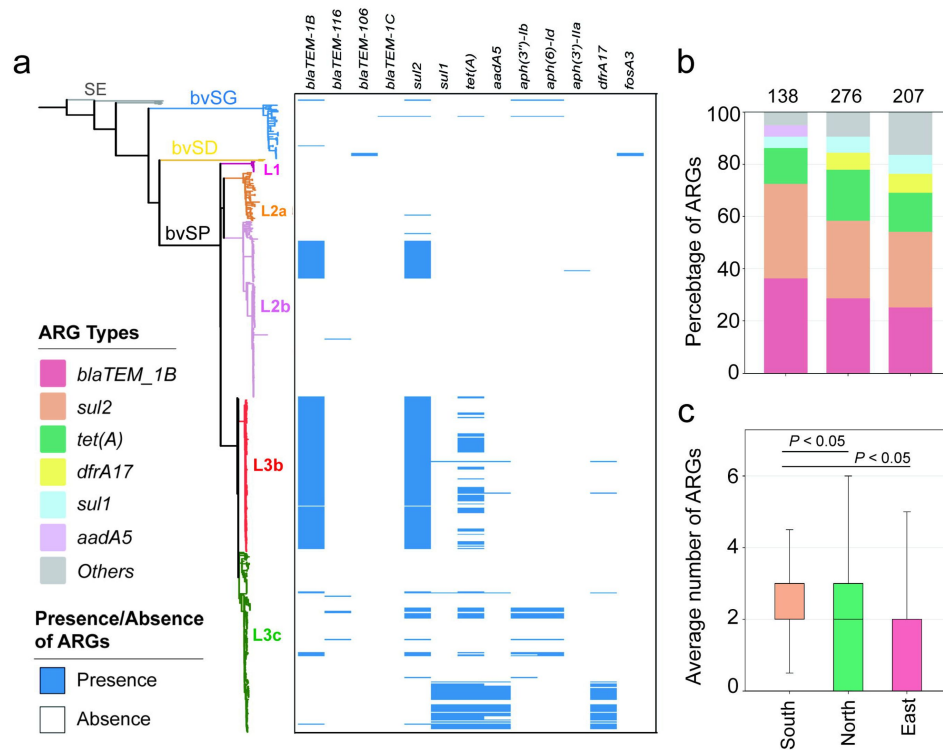


Figure 4.

The ARGs carried by *S. Gallinarum*.

(a) The phylogenetic tree was constructed using cgSNPs, revealing three *S. Gallinarum* biovars: *S. Gallinarum* biovar Pullorum (bvSP), *S. Gallinarum* biovar Duisburg (bvSD), and *S. Gallinarum* biovar Gallinarum (bvSG). Additionally, *Salmonella serovar* Enteritidis (SE.) is represented by a gray line. Further, bvSP can be subdivided into five lineages: fuchsia (L1), orange (L2a), pink (L2b), red (L3b), and green (L3c). The heatmap on the right indicates the resistome carried by the corresponding *Salmonella*. (b) The dominant resistome types in different regions of China. The y-axis represents the percentage of each dominant resistome. The total sample size for each bar is indicated at the top. (c) The average number of resistome carried by bvSP is from different regions of China.

Four categories of the mobilome—prophage, plasmid, transposon, and integron—were characterized, revealing strong lineage-specific patterns (S9a Fig, Supplementary Table 5). In silico analyses revealed prophages emerge as the most diverse mobilome type, with *Entero_mEp237*, *Escher_500465_1*, and *Klebsi_4LV2017* being more likely associated with L1, L2a, and L2b, respectively, while the carrier percentage of *Escher_pro483* is significantly higher among L3c. Interestingly, *Salmon_SJ46*, *Gifsy_2*, *Escher_500465_2*, and *Shigel_SfIV* were consistently observed across all bvSP lineages and were found to be duplicated on the chromosome. For plasmids, *IncFII(S)* was predominantly carried across all bvSP lineages (520/528, 98.5%) (S9b Fig). Notably, *ColpVC* had the lowest carriage percentage among L1 (2/10, 20%), whereas *IncX1* was explicitly associated with L3c (140/141, 99.3%).

The diversity of transposons and integrons is comparatively lower in bvSP. Specifically, we identified seven types of transposons and four types of integrons. Transposons were found to be more abundant, comprising a total of 255 in 528 bvSP. Among them, *Tn801* (185/528, 35%) had the highest carriage rate, followed by *Tn1721* (48/528, 9.1%). Interestingly, L3c is more likely to carry *Tn801* and *Tn1721* transposons than other lineages. In contrast, a total of 42 integrons were identified among the 528 bvSP, all of which belong to class I: *In498* (n=22), *In1440* (n=14), *In473* (n=4), and *In282* (n=2). The integrons also exhibited a lineage-preferential distribution, predominantly carried by L3b.

From a geographic distribution perspective, bvSP from northern and southern China carried the most extensive mobilome, followed by those from eastern China. Interestingly, bvSP isolated from northern China typically have more diverse mobilome types (S10 Fig).

Plasmid and transposon guide resistome geo-temporal dissemination

Variations in regional antimicrobial use may result in uneven pressure for selecting AMR. The mobilome is considered the primary reservoir for spreading resistome, and a consistent trend between the resistome and the mobilome has been observed across different lineages, from L1-L3c. We observed an overall gradual rise in the resistome quantity carried by bvSP across various lineages, correlating with the total mobilome content (S11 Fig). Furthermore, we investigated the interplay between particular mobile elements and resistome types in bvSP.

In bvSP, the predominant resistome is mainly associated with certain types of mobilome (**Fig 5a**). Specifically, regarding *bla*_{TEM-1B}, *tet(A)*, and *sul1*, which are highly prevalent among bvSP, we found most *bla*_{TEM-1B} were carried by the transposon *Tn801* and *Tn1721*, the plasmid *IncX1*, and the prophage SJ46. Additionally, *tet(A)* was primarily associated with the transposon *Tn1721* and the plasmid *IncX1*. Integrons also facilitated the dissemination of the resistome. For instance, *In498* and *In1440* facilitate the dissemination of *sul1*, leading to quinolone resistance. Our results indicate that plasmids and transposons are the predominant reservoirs for the resistome in bvSP, with prophages and integrons following closely behind (**Fig 5b**). Interestingly, we observed that the primary reservoir for the resistome shows regional variations. In the southern and northern regions of China, the primary reservoirs for *bla*_{TEM-1B} are *Tn801*, *Tn1721*, *IncX1*, and prophage SJ46. However, *Tn1721* was less frequently found in bvSP, which carries *bla*_{TEM-1B} from the eastern region. Similarly, the reservoir for *tet(A)* exhibits greater diversity in the east region but less diversity in the southern region (**Fig 5c-e**).

Horizontal transfer of resistome occurs widely in localized bvSP

Horizontal transfer of the resistome facilitates the acquisition of AMR among bacteria, which may record the distinct acquisition event in the bacterial genome. To compare these events in a geographic manner, we further investigated the HGT frequency of each ARG carried by bvSP isolated from China and explored the HGT frequency of resistome between three defined regions.

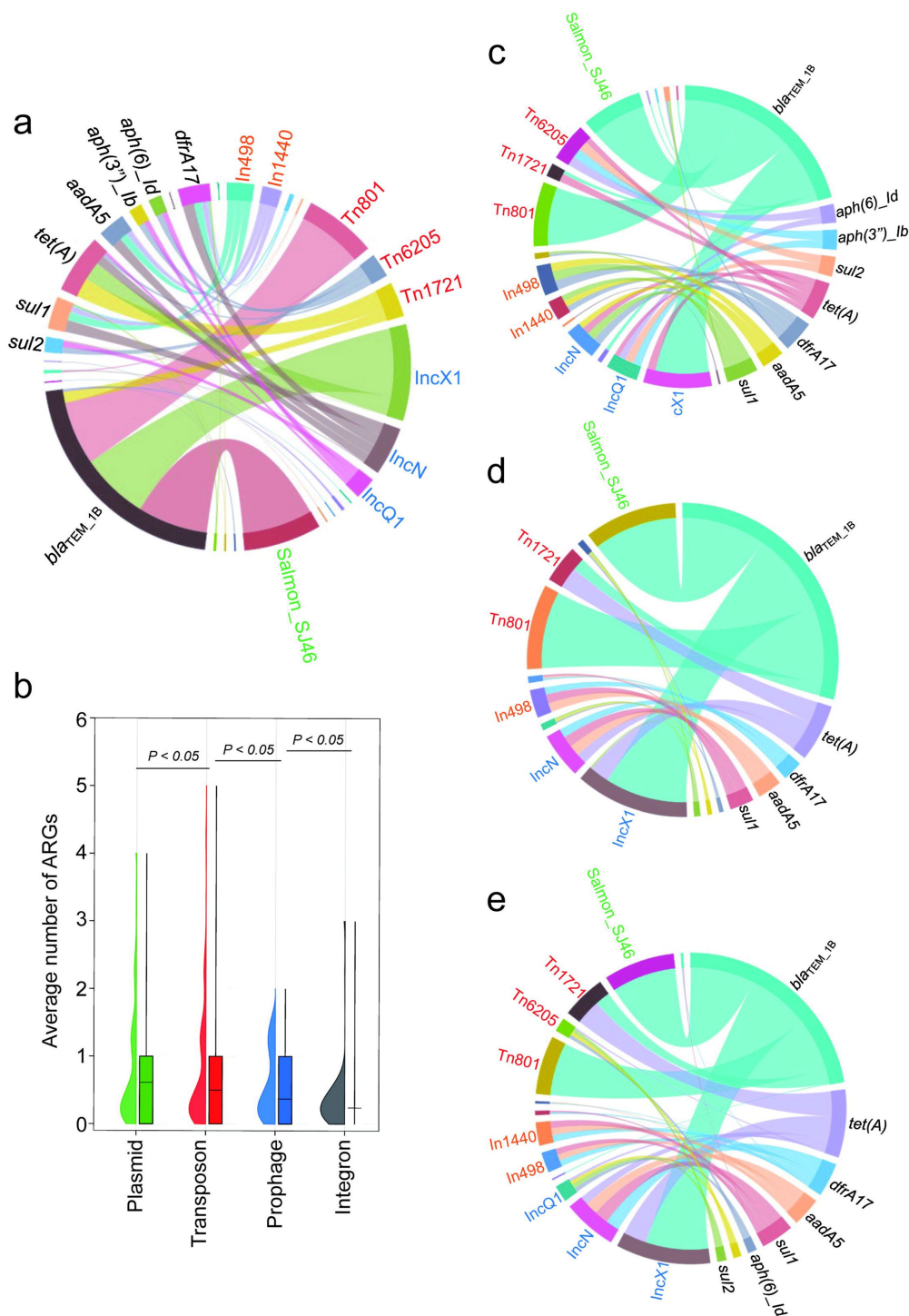


Figure 5.

The primary source of resistome is carried by distinct mobilome.

Different font colours denote various mobilome types. Specifically, orange signifies Integrons, red denotes transposons, blue represents plasmids, and green indicates prophages. A black font is utilized to distinguish the categories of resistome. The connecting line between the resistome and the mobilome represents the potential carrying relationship. (a) The mobilome-carried ARGs among *S. Gallinarum* biovar Pullorum (bvSP). (b) The average number of ARGs carried by the four MGEs in bvSP. The unpaired *t*-test was used to compare the differences between isolates, with $P < 0.05$ considered statistically significant. (c-e) The mobilome-carried ARGs among bvSP isolated from the eastern, southern, and northern regions of China.

Potentially horizontally transferred ARGs were defined as those with perfect identity (100% identity and 100% coverage) and were located on MGEs across different strains (Fig. 6a). We first categorized a total of 621 ARGs carried by 436 bvSP isolates collected in China and found that 415 ARGs were located on MGEs (Supplementary Table 9). After excluding the ARGs not associated with MGEs, our findings reveal that horizontal gene transfer of ARGs is widespread among Chinese bvSP isolates, with an overall transfer rate of 92%. Specifically, 50% of the ARGs exhibited an HGT frequency of 100%, indicating that these ARGs might undergo extensive frequent horizontal transfer events (Fig. 6b). It is noteworthy that certain resistance genes, such as *tet(A)*, *aph(3'')-Ib*, and *aph(6)-Id*, appear to be less susceptible to horizontal transfer.

However, different regions generally exhibited a considerable difference in resistome HGT frequency. Overall, bvSP from the southern areas in China showed the highest HGT frequency (HGT frequency=95%). The HGT frequencies for bvSP within the eastern and northern regions of China are lower, at 92% and 91%, respectively (Fig. 6c). For specific ARG type, we found *tet(A)* is more prone to horizontal transfer in the southern region, and this proportion was considerably lower in the eastern region. Interestingly, certain ARGs such as *aph(6)-Id*, undergo horizontal transfer only within the eastern and northern regions of China (Fig. 6d). Notably, as a localized transmission pathogen, resistome carried by bvSP exhibited a dynamic potential among inter-regional and local demographic transmission, especially from northern region to southern region (HGT frequency=93%) (Fig. 6e, Supplementary Table 7).

Discussion

Over the past decades, the emergence of AMR has garnered significant attention from both public health agencies and the poultry industry (25). Region-associated pressures shaped the dissemination patterns of bacterial pathogens, transitioning from pandemic to endemic, and led to dynamic changes in the resistome. Here, utilizing *S. Gallinarum* as a prototype for an endemic pathogen that once spread globally, and assembling the most comprehensive *S. Gallinarum* WGS database, we have discovered that region-associated pressures can influence the AMR risks of *S. Gallinarum* within lineage- and region-specific patterns. These pressures might also influence lineage-distinct evolutionary structures and transmission histories of *S. Gallinarum*.

The biovar types of *S. Gallinarum* have been well-defined as bvSP, bvSG, and bvSD historically (26). Among these, bvSP can be further subdivided into five lineages (L1, L2a, L2b, L3b, and L3c) using hierarchical Bayesian analysis. Different sublineages exhibited preferential geographic distribution, with L2b and L3b of bvSP being predominant global lineage types with a high risk of AMR. The historical geographical transmission was verified using a spatiotemporal Bayesian framework. The result shows that L3b was initially spread from China to Europe in the 18th-19th century, which may be associated with the European hen fever event in the mid-19th century (27). L2b, on the other hand, appears to have spread to Europe via South America, potentially contributing to the prevalence of bvSP in the United States. Considering the economic losses caused by bvSP, the United States, Europe, and other industrialized countries implemented eradication programs in the mid-19th century, thus eliminating the risk of pullorum disease. Implementing similar measures is challenging due to China's vast geographical area and various economic factors, resulting in bvSP becoming an endemic pathogen, mainly in China.

Additionally, forty-five bvSP was isolated from Taishun and Yueqing, to explore the transmission pattern of bvSP in China. The cgSNP distances between bvSP isolates from Taishun and Yueqing, showed significant diversity, with cgSNP distances of less than 100 isolate pairs observed exclusively within a single region. The transmission history of 45 global bvSP isolates revealed that most transmission events occurred within a single region. Additionally, the number of transmission events decreased rapidly with increasing distance, confirming that bvSP in China belongs to a highly localized pathogen.

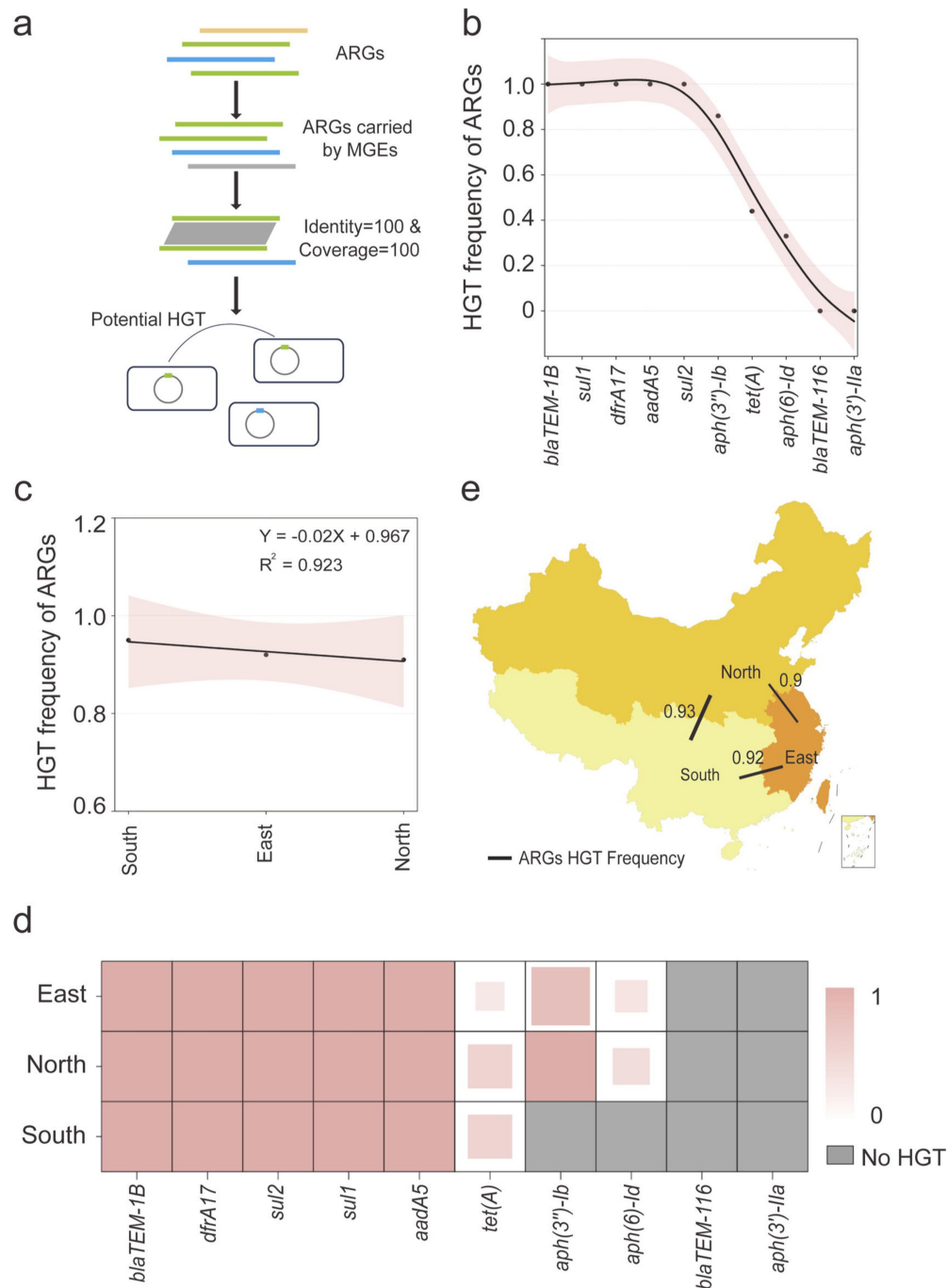


Figure 6.

The HGT frequency of the resistome among *S. Gallinarum* biovar *Pullorum* (bvSP) isolated from China.

(a) Workflow for identification of horizontally transferred ARGs in *Salmonella*. (b) The x-axis represents the resistome of bvSP, while the y-axis represents the corresponding levels of HGT frequency. (c) The overall resistome HGT frequency level of bvSP isolated from the various regions of China. (d) The HGT frequency level of specific ARGs carried by bvSP isolated from various regions of China. Deeper colours mean higher HGT frequency. (e) The frequency of horizontal retransmission of the resistome between different regions of China. A higher value indicates more frequent transfer events of resistome between two regions. The colors on the map represent the number of bvSP isolates in each region. Darker colors indicate a higher number of bvSP isolates in that area.

Humanity appeared powerless against epidemics until the advent of antimicrobials, which are still the primary agents used to combat *S. Gallinarum* infection (28 [↗](#), 29 [↗](#)). As a manifestation of localization-related forces, the emergence of AMR allows pathogens to acquire additional fitness benefits (30 [↗](#)).

Our findings further indicate that the mobilome facilitates the acquisition of AMR among bvSP, mainly through plasmids and transposons, such as *IncX1*, *IncQ1*, and *IncN* type plasmids, as well as *Tn801*, *Tn6205*, and *Tn721* type transposons. A survey conducted in China between 1962 and 2010 documented the association of class 1 integrons with AMR among bvSP (31 [↗](#)). Our study further demonstrates that *In498*, *In1440*, *In473*, and *In282*, all belonging to class 1 integrons, play a significant role in carrying ARGs associated with sulfonamides, and trimethoprim. However, compared to other *Salmonella* serovars (32 [↗](#)–34 [↗](#)), the level of resistome in bvSP remains lower.

Considering lineage prospects, recently emerged lineages present a higher risk of AMR, with a lineage-specific distribution of resistome. In L2b and L3c, the most prevalent groups of ARGs include *bla*_{TEM-1b} and *sul2*. L3c, which exhibits the highest resistome richness with *sul1*, *tet(A)*, *aadA5*, and *dhfrA17*, has a higher likelihood of MDR. However, the transmission routes of resistome showed a high degree of consistency across continents. Globally, a small proportion of resistome carriers belonging to L2b and L3b were observed in the Americas and Europe, while isolates from Asia exhibited a notable rise in resistome carriage. Therefore, we speculate that the mobilome in bvSP may have been acquired locally in Asia, possibly due to increased antimicrobial usage and an intensive industrialized poultry farming pattern within the regional poultry industry (35 [↗](#)). The L3c, being a local Asian lineage and more prone to acquiring additional resistome locally, provided further support.

In China, the endemic region of Asia, an unexpected rising trend has been observed from the 1970s to the 2020s. However, the risk of AMR varies among distinct regions (36 [↗](#)–39 [↗](#)). Overall, the average number of ARGs carried by bvSP isolated from eastern China was the lowest, while it was higher among bvSP isolated from northern and southern China. This variation may be influenced by differences in the economy, climate, or farming practices among different regions. Interestingly, the widespread horizontal transfer of the resistome across regions seems to indicate that the resistome has a different mode of transmission compared to its pathogenic bacteria. Specifically, as a localized pathogen, our results show a low frequency of cross-regional transmission of bvSP (as shown in S5 Fig), but up to 90% of ARGs are transmitted cross-regionally. This risk has not been previously observed in *S. Gallinarum*, suggesting that potential intermediate hosts may play a key role.

In summary, the findings of this study highlight that *S. Gallinarum* remains a significant concern in developing countries, particularly in China. Compared to other regions, *S. Gallinarum* in China poses a notably higher risk of AMR, necessitating the development of additional therapies, i.e. vaccine, probiotics, bacteriophage therapy in response to the government's policy aimed at reducing antimicrobial use (40 [↗](#)). Furthermore, given the dynamic nature of *S. Gallinarum* risks across different regions, it is crucial to prioritize continuous monitoring in key areas, particularly in China's southern regions where the extensive poultry farming is located. Lastly, from a One-Health perspective, controlling AMR in *S. Gallinarum* should not solely focus on local farming environments, with improved overall welfare on poultry and farming style. The breeding pyramid of industrialized poultry production should be targeted on the top, with enhanced and accurate detection techniques (41 [↗](#)). More importantly, comprehensive efforts should be made to reduce antimicrobial usage overall and mitigate potential AMR transmission from environmental sources or other hosts (42 [↗](#)–44 [↗](#)).

However, the current study has some limitations. Firstly, despite assembling the most comprehensive WGS database for *S. Gallinarum* from public and laboratory sources, there are still biases in the examined collection. The majority (438/580) of *S. Gallinarum* samples were collected

from China, possibly since the WGS is a technology that only became widely available in the 21st century. This makes it impractical to sequence it on a large scale in the 20th century, when *S. Gallinarum* caused a global pandemic. So, we suspect that human intervention in the development of this epidemic is the main driving force behind the fact that most of the strains in the data set originated in China. In our future work, we aim to actively gather more data to minimize potential biases within our dataset, thereby improving the robustness and generalizability of our findings. Secondly, *in silico* analysis relies heavily on sophisticated, continuously updated databases. Although we utilized the most up-to-date databases, the exact prevalence of mobilome genetic elements (MGEs) and ARGs may be underestimated. Nevertheless, by using the most extensive global WGS data from *S. Gallinarum*, we elucidate a dynamic resistome evolution framework in a single bacterial pathogen from pandemic to endemic, which is hallmarked by a gradual increase in antimicrobial resistance risks and a stepwise resistome adaptation. The insights gained in this study will be helpful in further prevention and surveillance of the AMR risks of localized pathogens.

Material and Methods

Ethics statements

The protocols used for animal studies were approved by the Committee of the Laboratory Animal Center of Zhejiang University (ZJU20190094; ZJU20220295).

Bacterial isolates

All 734 samples of dead chicken embryos aged 19 to 20 days were collected from Taishun and Yueqing in Zhejiang Province, China. After a thorough autopsy, the liver, intestines, and spleen were extracted and added separately into 2 mL centrifuge tubes containing 1 mL PBS. The organs were then homogenized by grinding. In the initial enrichment phase, we utilized Buffered Peptone Water (BPW, Haibo Biotechnology Co, Qingdao, China), employing a 1:9 dilution ratio (sample in PBS: BPW). Subsequently, the composite was incubated at 37°C for 16–18 hours in a rotary incubator set to 180 rpm. For selective enrichment, Tetrathionate Broth Base (TTB, Land Bridge Biotechnology Co, Beijing, China), fortified with iodine solution and brilliant green solution (both from Land Bridge Biotechnology Co, Beijing, China), was employed at a ratio of 1:10 (sample in BPW: TTB). The mixture was subjected to incubation at 42°C for a duration ranging between 22 and 26 hours within a rotary incubator set at 180 rpm. Isolated *Salmonella* colonies from positive samples were obtained by sub-culturing selectively enriched samples on Xylose Lysine Deoxycholate (XLD, Land Bridge Technology Co, Beijing, China) agar, followed by an 18–22-hour incubation at 37°C. Typical and pure colonies were selected after sub-culturing on XLD agar and transferred into Luria-Bertani (LB) broth. Finally, the transferred bacterial culture underwent an additional incubation for 18–22 hours at 37°C in a rotary incubator operating at 180 rpm.

DNA extraction and genomic assembly

The DNA of forty-five isolates was extracted using the Vazyme Fastpure[®] Bacteria DNA Isolation Mini Kit (Vazyme Biotech Co., Ltd.), which was then quantified using the NanoDrop1000 system (Thermo Fisher Scientific, USA). Subsequently, DNA libraries were constructed and subjected to sequencing using the Illumina Novaseq 6000 platform (Beijing Novogene Co., Ltd). Assembly of genome sequences was performed by SPAdes ([45](https://doi.org/10.1093/bioinformatics/btu160)) v3.12.0 with default parameters.

Global dataset assembly

To provide global context, we expanded the dataset by including 540 additional WGS genome data, consisting of bvSP (n=483), bvSG (n=50), bvSD (n=2), and *Salmonella enterica* serovar Enteritidis (n=5). The inclusion of five strains of *Salmonella enterica* serovar Enteritidis enhanced

evolutionary analyses. Notably, in this collection of 540 genomic data from public databases, 325 sequences were previously preserved in our laboratory.

All WGS data passed strict quality control according to criteria set by the European Reference Laboratory (46). Genomic data exceeding 500 contigs and having an N50 of less than 30,000 were excluded. Lastly, the bacterial species associated with each genomic data set was confirmed by utilizing KmerFinder (47) v3.2. Finally, the assembled database includes a total of 585 high-quality sequences.

Genotyping and phylogenetic analyses

Snippy v.4.4.5 was used to identify cgSNPs, with the complete genome of R51 used as a reference. The optimal model (TVM+F) was determined and employed to construct the maximum-likelihood phylogenetic tree using IQ-TREE v.1.6 (48). The final phylogenetic tree was annotated and visualized using iTOL v.6.0 (49).

The global assembly of *S. Gallinarum* underwent a population structure analysis using the RhierBAPS v1.1.4 (50), based on the Hierarchical Bayesian clustering (BHC) algorithm and employing cgSNPs as input. The calculation of population structure utilized R package v.4.3.1, supplemented by the R packages “phytools” v.2.0.3, “ape” v.5.7.1, and “rhierbaps” v.1.1.4. Default parameters were employed for all population structure analysis.

Temporal and phylogeographical analysis

To examine the emergence and geographical transfers, we first utilized Gubbins v.2.3 (51) to eliminate the recombination regions for each lineages (S12 Fig). Then, TreeTime (52) was used to assess the temporal structure. This was accomplished through regression analysis of the root-to-tip branch distances within the maximum likelihood tree, considering the sampling date as a variable. Then, the strains within the dataset underwent phylogeographical reconstruction via Bayesian Evolutionary Analysis of Sampled Trees (BEAST) (53) v.2.5. To determine the optimal model for running BEAST, we tested a total of six combinations in the initial phase of our study. These combinations included the strict clock, relaxed lognormal clock, and three population models (Bayesian SkyGrid, Bayesian Skyline, and Constant Size). Before conducting the full BEAST analysis, we evaluated each combination using a Markov Chain Monte Carlo (MCMC) analysis with a total chain length of 100 million and sampling every 10,000 iterations. We then summarized the results using NSLogAnalyser and determined the optimal model based on the marginal likelihood value for each combination. The results indicated that the model incorporating the Bayesian Skyline, and the relaxed lognormal clock yielded the highest marginal likelihood value in our sample. Consequently, we proceeded to perform a time-calibrated Bayesian phylogenetic inference analysis for each lineage. The following settings were configured: the “GTR” substitution model, “4 gamma categories”, the “Relaxed Clock Log Normal” model, the “Coalescent Bayesian Skyline” tree prior. Convergence was assessed using Tracer, with all parameter effective sampling sizes (ESS) exceeding 200. Maximum clade credibility trees were generated using TreeAnnotator v.2.6.7. Finally, key divergence time points (with 95% credible intervals) were estimated and the phylogenetic tree was visualized using Figtree v.1.4.3.

SNP distance-based *S. Gallinarum* geographical tracing

To understand potential transmission events, we calculated the cgSNPs distance between distinct bacterial strains employing the SNP-dists. We estimated the overall evolutionary rate of the *S. Gallinarum* using BEAST. We applied the methodology described previously (54). The numbers of SNPs per year were determined by multiplying the evolutionary rates estimated with BEAST by the number of core SNP sites identified in the alignments. We hypothesize that a slower evolutionary rate in bacteria typically requires a lower SNP threshold when tracing transmission events using SNP distance analysis. Previous research found an average evolutionary rate of 1.97

SNPs per year (95% HPD, 0.48 to 4.61) across 22 different *Salmonella* serotypes. Our updated BEAST estimation for the evolutionary rate of *S. Gallinarum* suggests it is approximately 0.74 SNPs per year (95% HPD, 0.42 to 1.06). Based on these findings, as well as our previous experience with similar studies (55), we set a threshold of 5 SNPs. Additionally, the geographic location was considered to enhance the precision of speculation regarding potential transmission sites and transmission direction.

MGE detecting

Four types of MGEs were detected: plasmids, transposons, integrons, and prophages. Plasmids were detected using Abricate v.1.0.1 with the PlasmidFinder (56) database. Only plasmids with a similarity of more than 90% and a coverage over 95% were identified. BacAnt (57) was used to detect integrons and transposons in the genomes. Only integrons or transposons with a similarity of more than 60% and a coverage greater than 60% will be identified. The prophages were detected using the Phaster pipeline. The genomic data were split into two temporary databases based on the number of contigs, one dataset containing single contig files and the other containing multiple contig files. The two databases were imported into the Phaster pipeline separately with default parameters.

Determination of the potential MGE carried ARG and horizontal ARG transfer frequency

We have devised a pipeline to discern the coexistence of MGEs and ARGs. For plasmids and prophages, only ARGs situated within MGE regions are deemed MGE-carried. Regarding transposons and integrons, the search region was expanded to 5 kilobases upstream and downstream of ARGs, accounting for potential splicing errors. In this study, HGT events involving ARGs were defined as instances where ARGs exhibited perfect identity (100% coverage and 100% identity) and were located on MGEs across different strains. In a previous study (58), horizontal transfer of ARGs was identified based on gene regions with 100% coverage and over 99% identity. We believe that these stricter criteria in our pipeline more accurately reflect the transfer of ARGs within the *S. Gallinarum* population. HGT frequency was utilized to evaluate the extent of horizontal transfer of ARGs as follows:

$$HGT\ frequency = \frac{\text{Horizontal ARG transfer Count}}{\text{Total dissemination count}}$$

The pipeline developed in this study accepts Resfinder or Resistance Gene Identifier (RGI) results as input. Specific code and examples are uploaded to: https://github.com/tjiaa/Cal_HGT_Frequency/tree/main.

Genomic data analysis

The serovars of all WGS genome data were confirmed through SISTR (59) v.1.1 and Seqsero2 (60). Multilocus Sequence Typing was carried out using MLST v.2.22 with the *senterica_achtman_2* scheme. ARGs were detected by ResFinder (61). Both similarity and coverage were set to a minimum value of 90. The invasiveness index was calculated using methods previously reported (62). Specifically, *Salmonella*'s ability to cause intestinal or extraintestinal infections in hosts is related to the degree of genome degradation. We evaluated the potential for extraintestinal infection by 45 newly isolated *S. Gallinarum* strains (L2b and L3b) using a model that quantitatively assesses genome degradation. We analyzed each sample using the 196 top predictor genes for measuring the invasiveness of *S. Gallinarum*, employing a machine-learning approach that utilizes a random forest classifier and delta-bitscore functional variant-calling. This method evaluated the invasiveness of *S. Gallinarum* towards the host, and the distribution of invasiveness index values for each region was statistically tested using unpaired t-test. The code used for calculating the invasiveness index is available at https://github.com/Gardner-BinfLab/invasive_salmonella.

Additional information

Data availability

For the newly isolated 45 strains of *Salmonella* Gallinarum, genome data have been deposited in NCBI Sequence Read Archive (SRA) database. The “SRA Accession” for each strain are listed in Supplementary Table 1. Additionally, the genome data for the 540 publicly available genomes have been uploaded to Figureshare and can be accessed at: https://figshare.com/articles/journal_contribution/Genomic_data/26028592.

Code availability

The open-source software used in this study includes:

BacAnt (<https://github.com/xthua/bacant>),

Abricate (<https://github.com/tseemann/abricate>),

Snippy (<https://github.com/tseemann/snippy>),

SISTR (https://github.com/phac-nml/sistr_cmd),

MLST (<https://cge.food.dtu.dk/services/MLST/>)

SeqSero2 (<https://github.com/denglab/SeqSero2>),

IQtree (<https://github.com/iqtree/iqtree2>).

BEAST (<https://www.beast2.org>)

SNP-dists (<https://github.com/tseemann/snp-dists>)

Fast BAPS (<https://github.com/gtonkinhill/fastbaps>)

R (<https://www.r-project.org>)

Phaster (<http://phaster.ca>)

KmerFinder (<https://bitbucket.org/genomicepidemiology/kmerfinder>)

Treetime (<https://github.com/neherlab/treetime>)

Author contributions

M.Y., C.J. designed the study; finalized the manuscript;

C.J. collected isolates and clinical data, integrated the data, performed metadata and genomic analysis, prepared tables and figures; and wrote the draft manuscript;

C.H. performed metadata and genomic analysis; interpreted the data; finalized the manuscript;

X.Z., X.K., Q.C., Y.H. and Z.W. contributed to the data analysis;

C.H., M.Y. prepared tables and figures;

M.Y., C.H., A.S., Y.L., F.H., and C.J. finalized the manuscript;

M.Y. managed the project;

All the authors contributed to the article and approved the submitted version.

Declaration of Interests

None.

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Reviewer #1 (Public review):

Summary:

The investigators in this study analyzed the dataset assembly from 540 *Salmonella* isolates, and those from 45 recent isolates from Zhejiang University of China. The analysis and comparison of the resistome and mobilome of these isolates identified a significantly higher rate of cross-region dissemination compared to localized propagation. This study highlights the key role of the resistome in driving the transition and evolutionary history of *S. Gallinarum*.

Strengths:

The isolates included in this study were from 16 countries in the past century (1920 to 2023). While the study uses *S. Gallinarum* as the prototype, the conclusion from this work will likely apply to other *Salmonella* serotypes and other pathogens.

Weaknesses:

While the isolates came from 16 countries, most strains in this study were originally from China.

Comments on revisions:

This reviewer is happy with the detailed responses from the authors regarding revising this manuscript. I do not have further comments.

<https://doi.org/10.7554/eLife.101241.2.sa2>

Reviewer #2 (Public review):

Summary:

The authors sequence 45 new samples of *S. Gallinarum*, a commensal *Salmonella* found in chickens, which can sometimes cause disease. They combine these sequences with around 500 from public databases, determine the population structure of the pathogen, and coarse relationships of lineages with geography. The authors further investigate known anti-microbial genes found in these genomes, how they associate with each other, whether they have been horizontally transferred, and date the emergence of clades.

Strengths:

- It doesn't seem that much is known about this serovar, so publicly available new sequences from a high burden region are a valuable addition to the literature.
- Combining these sequences with publicly available sequences is a good way to better contextualise any findings.
- The genomic analyses have been greatly improved since the first version of the manuscript, and appropriately analyse the population and date emergence of clades.
- The SNP thresholds are contextualised in terms of evolutionary time.
- The importance and context of the findings are fairly well described.

Weaknesses:

- There are still a few issues with the genomic analyses, although they no longer undermine the main conclusions:

(1) Although the SNP distance is now considered in terms of time, the 5 SNP distance presented still represents ~7yrs evolution, so it is unlikely to be a transmission event, as described. It would be better to use a much lower threshold or describe the interpretation of these clusters more clearly. Bringing in epidemiological evidence or external references on the likely time interval between transmissions would be helpful.

(2) The HGT definition has not fundamentally been changed and therefore still has some issues, mainly that vertical evolution is still not systematically controlled for. Using a 5kb window is not sufficient, as LD may extend across the entire genome. As the authors have now run gubbins correctly, they could use the results from this existing analysis to find recent HGT. To definite mobilisation, perhaps a standard pipeline such (e.g. <https://github.com/EBI-Metagenomics/mobilome-annotation-pipeline>) would be more convincing.

(3) The invasiveness index is better described, but the authors still did not provide convincing evidence that the small difference is actually biologically meaningful (there was no statistical difference between the two strains provided in response Figure 6). What do other *Salmonella* papers using this approach find, and can their links be brought in? If there is still no good evidence, a better description of this difference would help make the conclusions better supported.

In summary, the analysis is broadly well described and feels appropriate. Some of the conclusions are still not fully supported, although the main points and context of the paper now appear sound.

<https://doi.org/10.7554/eLife.101241.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

The investigators in this study analyzed the dataset assembly from 540 Salmonella isolates, and those from 45 recent isolates from Zhejiang University of China. The analysis and comparison of the resistome and mobilome of these isolates identified a significantly higher rate of cross-region dissemination compared to localized propagation. This study highlights the key role of the resistome in driving the transition and evolutionary

Thank you for summarizing our work. According to your comments, we carefully considered and responded to them and made corresponding revisions to the text. Additionally, to fully contextualize the background knowledge and clarify the major points in this study, we add some references.

Upon further review of our initial manuscript, we realized that the original submission did not strictly follow the lineage order proposed by Zhou et al. (Natl Sci Rev. 2023 Sep 2;10(10):nwad228). To avoid confusion and keep the uniform knowledge in the typing system, we have adjusted the lineage nomenclature along the revised manuscript to reflect the corrected order as follows:

Author response table 1:

To ensure consistency with previous studies, we have revised the nomenclature for the different lineages of bvSP.

Lineage name modification
L1 > L1
L2a > L2a
L2b > L2b
L3b > L3c
L3c > L3b

Strengths:

The isolates included in this study were from 16 countries in the past century (1920 to 2023). While the study uses S. Gallinarum as the prototype, the conclusion from this work will likely apply to other Salmonella serotypes and other pathogens.

Thanks for the constructive comments and the positive reception of the manuscript.

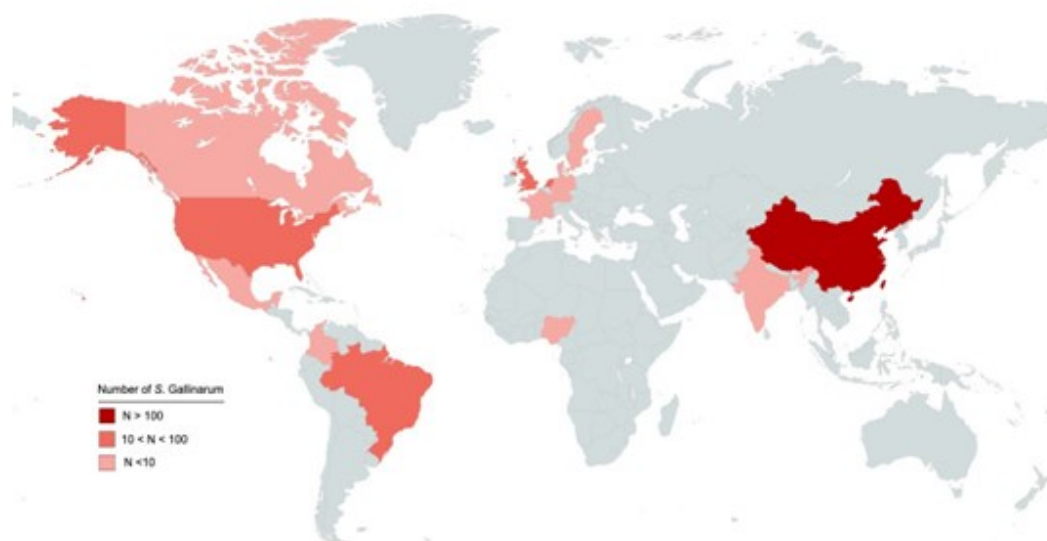
Weaknesses:

While the isolates came from 16 countries, most strains in this study were originally from China.

We appreciate the reviewer's observation regarding the sampling distribution of isolates in this study. We acknowledge that while the isolates were collected from 15 different countries, with a significant proportion originated from China (Author response image 1). This focus is due to several reasons:

Author response image 1.

Geographic distribution of 580 *S. Gallinarum*. Different colors indicate the countries of origin for the 580 *S. Gallinarum* strains in the dataset. Darker shades represent higher numbers of strains.

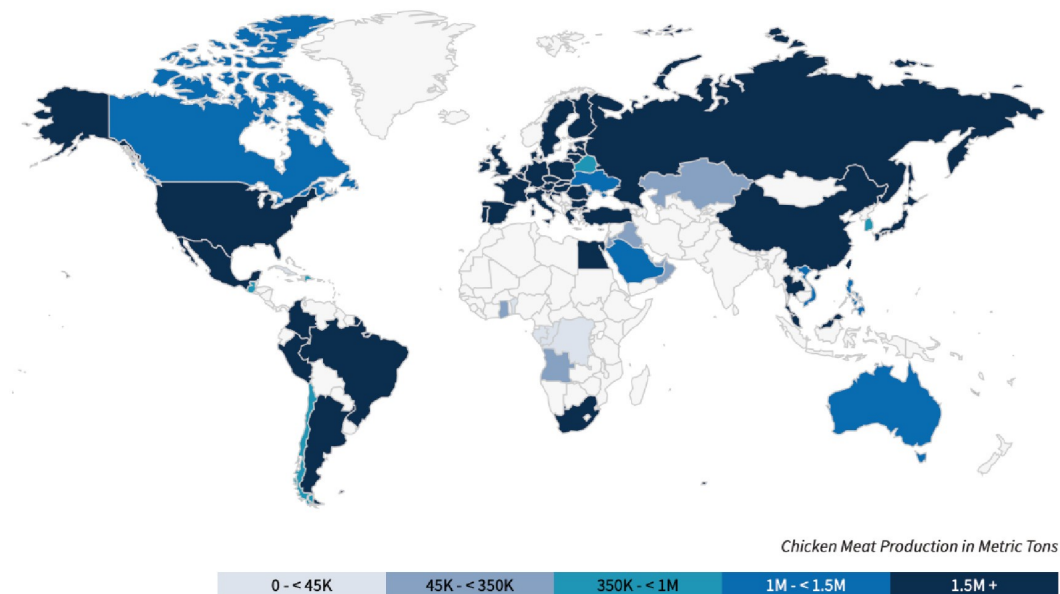


(1) As once a globally prevalent pathogen across the 20th century, *S. Gallinarum* was listed by the World Organization for Animal Health (WOAH) due to its economic importance. After 30 years of implementation of the National Poultry Improvement Plan in the US, it was almost eradicated in high-income countries, and interestingly, it became an endemic pathogen with sporadic outbreaks in most low- or middle-income countries like China and Brazil. Given the vast expanse of China's land area and the country's economic factors, implementing the same measures remains challenging.

(2) *S. Gallinarum* is an avian-specific pathogen, particularly affecting chickens, and its distribution is closely linked to chicken meat production in different countries. There are more frequent reports of fowl typhoid in some high chicken-producing developing countries. Data from the United States Department of Agriculture (USDA) on annual chicken meat production for 2023/2024 show that the global distribution of *S. Gallinarum* aligns closely with the overall chicken meat production of these countries (<https://fas.usda.gov/data/production/commodity/0115000>).

Author response image 2.

The United States Department of Agriculture (USDA) data on annual chicken meat production for 2023/2024 across different countries globally.

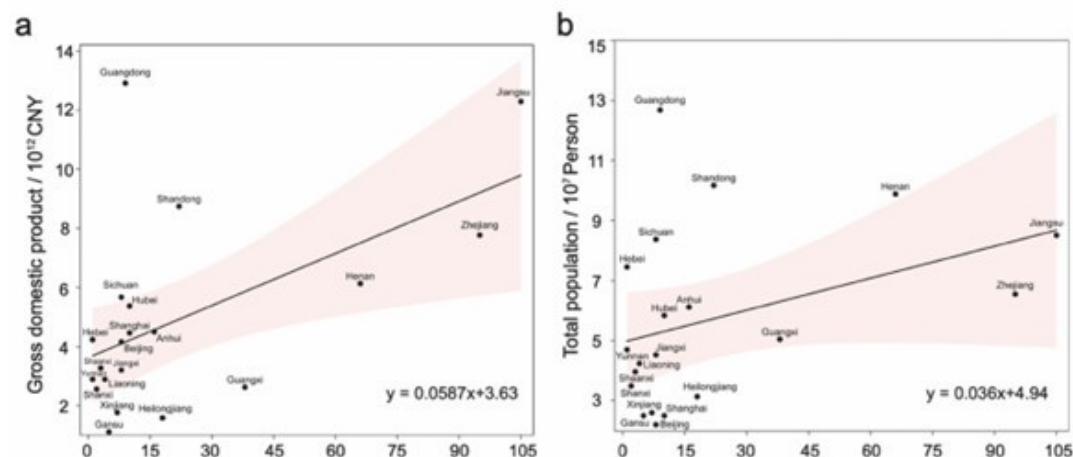


(3) Our primary objective was to investigate the localized resistome adaptation of *S. Gallinarum* in regions. Being a region with significant disease burden, China has reported numerous outbreaks (Sci Data. 2022 Aug 13;9(1):495; Sci Data. 2024 Feb 27;11(1):244) and a high AMR prevalence of this serovar (Natl Sci Rev. 2023 Sep 2;10(10):nwad228; mSystems. 2023 Dec 21;8(6):e0088323), making it an excellent example for understanding localized resistance mechanisms.

(4) As China is the primary country of origin for the strains in this study, it is necessary to ensure that the strains from China are consistent with the local geographic characteristics of the country. Therefore, we conducted a correlation analysis between the number of strains from different provinces in China and the total GDP/population size of those provinces (Author response image 3). The results show that most points fall within the 95% confidence interval of the regression line. Although some points exhibit relative unbalance in the number of *S. Gallinarum* strains, most data points for these regions have a small sample size ($n < 15$). Overall, we found that the prevalence of *S. Gallinarum* in different regions of China is consistent with the overall nationwide trend.

Author response image 3.

Correlation analysis between the number of *S. Gallinarum* collected from different provinces in China and the total GDP/population size. The figure depicts a series of points representing individual provinces. The x-axis indicates the number of *S. Gallinarum* included in the dataset, while the y-axis displays the values for total GDP and total population size, respectively.



Nevertheless, a search of nearly a decade of literature on PubMed and a summary of the *S. Gallinarum* genome available on public databases indicate that the dataset used is the most complete. Furthermore, focusing on a specific region within China allowed us to conduct a detailed and thorough analysis. However, we highly agree that expanding the study to include more isolates from other countries would enhance the generalizability of our findings, and we are actively collecting additional *S. Gallinarum* genome data. In the revised manuscript, we have further emphasized the limitations as follow:

Lines 427-429: “However, the current study has some limitations. Firstly, despite assembling the most comprehensive WGS database for *S. Gallinarum* from public and laboratory sources, there are still biases in the examined collection. The majority (438/580) of *S. Gallinarum* samples were collected from China, possibly since the WGS is a technology that only became widely available in the 21st century. This makes it impractical to sequence it on a large scale in the 20th century, when *S. Gallinarum* caused a global pandemic. So, we suspect that human intervention in the development of this epidemic is the main driving force behind the fact that most of the strains in the data set originated in China. In our future work, we aim to actively gather more data to minimize potential biases within our dataset, thereby improving the robustness and generalizability of our findings.”

Reviewer #2 (Public review):

Summary:

The authors sequence 45 new samples of S. Gallinarum, a commensal Salmonella found in chickens, which can sometimes cause disease. They combine these sequences with around 500 from public databases, determine the population structure of the pathogen, and coarse relationships of lineages with geography. The authors further investigate known anti-microbial genes found in these genomes, how they associate with each other, whether they have been horizontally transferred, and date the emergence of clades.

Thank you for your constructive suggestions, which are valuable and highly beneficial for improving our paper. According to your comments, we carefully considered and responded to them and made corresponding revisions to the text. Furthermore, to fully contextualize the background knowledge and clarify the major points in this study, we add some references to support our findings and policy implications.

Upon further review of our initial manuscript, we realized that the original submission did not strictly follow the lineage order proposed by Zhou et al. (Natl Sci Rev. 2023 Sep 2;10(10):nwad228). To avoid confusion in the typing system, we have adjusted the lineage

nomenclature in the revised manuscript to reflect the corrected order (see Author response table 1).

Strengths:

- (1) It doesn't seem that much is known about this serovar, so publicly available new sequences from a high-burden region are a valuable addition to the literature.*
- (2) Combining these sequences with publicly available sequences is a good way to better contextualise any findings.*

Thank you so much for your thorough review and constructive comments on the manuscript.

Weaknesses:

There are many issues with the genomic analysis that undermine the conclusions, the major ones I identified being:

- (1) Recombination removal using gubbins was not presented fully anywhere. In this diversity of species, it is usually impossible to remove recombination in this way. A phylogeny with genetic scale and the gubbins results is needed. Critically, results on timing the emergence (fig2) depend on this, and cannot be trusted given the data presented.*

We sincerely thank you for pointing out this issue. In the original manuscript, we aimed to present different lineages of *S. Gallinarum* within a single phylogenetic tree constructed using BEAST. However, in the revised manuscript, we have addressed this issue by applying the approach recommended by Gubbins to remove recombination events for each lineage defined by FastBAPs. Additionally, to better illustrate the removal of recombination regions in the genome, we have included a figure generated by Gubbins (New Supplementary Figure 12).

Our results indicate that recombination events are relatively infrequent in Lineage 1, followed by Lineage 3, but occur more frequently in Lineage 2. In the revised manuscript, we have included additional descriptions in the Methods section to clarify this analysis. We hope these modifications adequately address the reviewer's concerns and enhance the trustworthiness of our findings.

- (2) The use of BEAST was also only briefly presented, but is the basis of a major conclusion of the paper. Plot S3 (root-to-tip regression) is unconvincing as a basis of this data fitting a molecular clock model. We would need more information on this analysis, including convergence and credible intervals.*

Thank you very much for raising this issue. We decided to reconduct separate BEAST analyses for each lineage, accurately presenting the evolutionary scale based on the abovementioned improvements. The implementation of individual lineage for BEAST analysis was conducted based on the following steps:

- (1) Using R51 as the reference, a reference-mapped multiple core-genome SNP sequence alignment was created, and recombination regions were detected and removed as described above.
- (2) TreeTime was used to assess the temporal structure by performing a regression analysis of the root-to-tip branch distances within the maximum likelihood tree, considering the sampling date as a variable (New Supplementary Figures 6). However, the root-to-tip regression analysis presented in New Supplementary Figures 6 was not intended as a basis

for selecting the best molecular clock model; its purpose was to clean the dataset with appropriate measurements.

(3) To determine the optimal model for running BEAST, we tested a total of six combinations in the initial phase of our study. These combinations included the strict clock, relaxed lognormal clock, and three population models (Bayesian SkyGrid, Bayesian Skyline, and Constant Size). Before conducting the complete BEAST analysis, we evaluated each combination using a Markov Chain Monte Carlo (MCMC) analysis with a total chain length of 100 million and sampling every 10,000 iterations. We then summarized the results using NSLogAnalyser and determined the optimal model based on the marginal likelihood value for each combination. The results indicated that the model incorporating the Bayesian Skyline and the relaxed lognormal clock yielded the highest marginal likelihood value in our sample. Then, we proceeded to perform a timecalibrated Bayesian phylogenetic inference analysis for each lineage. The following settings were configured: the "GTR" substitution model, "4 gamma categories", the "Relaxed Clock Log Normal" model, the "Coalescent Bayesian Skyline" tree prior, and an MCMC chain length of 100 million, with sampling every 10,000 iterations.

(4) Convergence was assessed using Tracer, with all parameter effective sampling sizes (ESS) exceeding 200. Maximum clade credibility trees were generated using TreeAnnotator. Finally, key divergence time points (with 95% credible intervals) were estimated, and the tree was visualized using FigTree.

For the key lineages, L2b and L3b (carrying the resistome, posing antimicrobial resistance (AMR) risks, and exhibiting intercontinental transmission events), we have redrawn Figure 2 based on the updated BEAST analysis results (New Figure 2). For L1, L2a, and L3c, we have added supplementary figures to provide a more detailed visualization of their respective BEAST analysis outcomes (New Supplementary Figures 3-5). The revised BEAST analysis indicates that the origin of L3b in China can be traced back to as early as 1683 (95% CI: 1608 to 1839). In contrast, the earliest possible origin of L2b in China dates back to 1880 (95% CI: 1838 to 1902). This indicates that the previous manuscript's assumption that L2b is an older lineage compared to L3b may be inaccurate.

Furthermore, In the revised manuscript, we specifically estimated the time points for the first intercontinental transmission events for the two major lineages, L2b and L3b. Our results indicate that L2b, likely underwent two major intercontinental transmission events. The first occurred around 1893 (95% CI: 1870 to 1918), with transmission from China to South America. The second major transmission event occurred in 1923 (95% CI: 1907 to 1940), involving the spread from South America to Europe. In contrast, the transmission pattern of L3b appears relatively more straightforward. Our findings show that L3b, an *S. Gallinarum* lineage originating in China, only underwent one intercontinental transmission event from China to Europe, likely occurring around 1790 (95% CI: 1661 to 1890) (New Supplementary Figure 7). Based on the more critical BEAST analysis for each lineage, we have revised the corresponding conclusions in the manuscript. We believe that the updated BEAST analysis, performed using a more accurate recombination removal approach, significantly enhances the rigor and credibility of our findings.

(3) Using a distance of 100 SNPs for a transmission is completely arbitrary. This would at least need to be justified in terms of the evolutionary rate and serial interval.

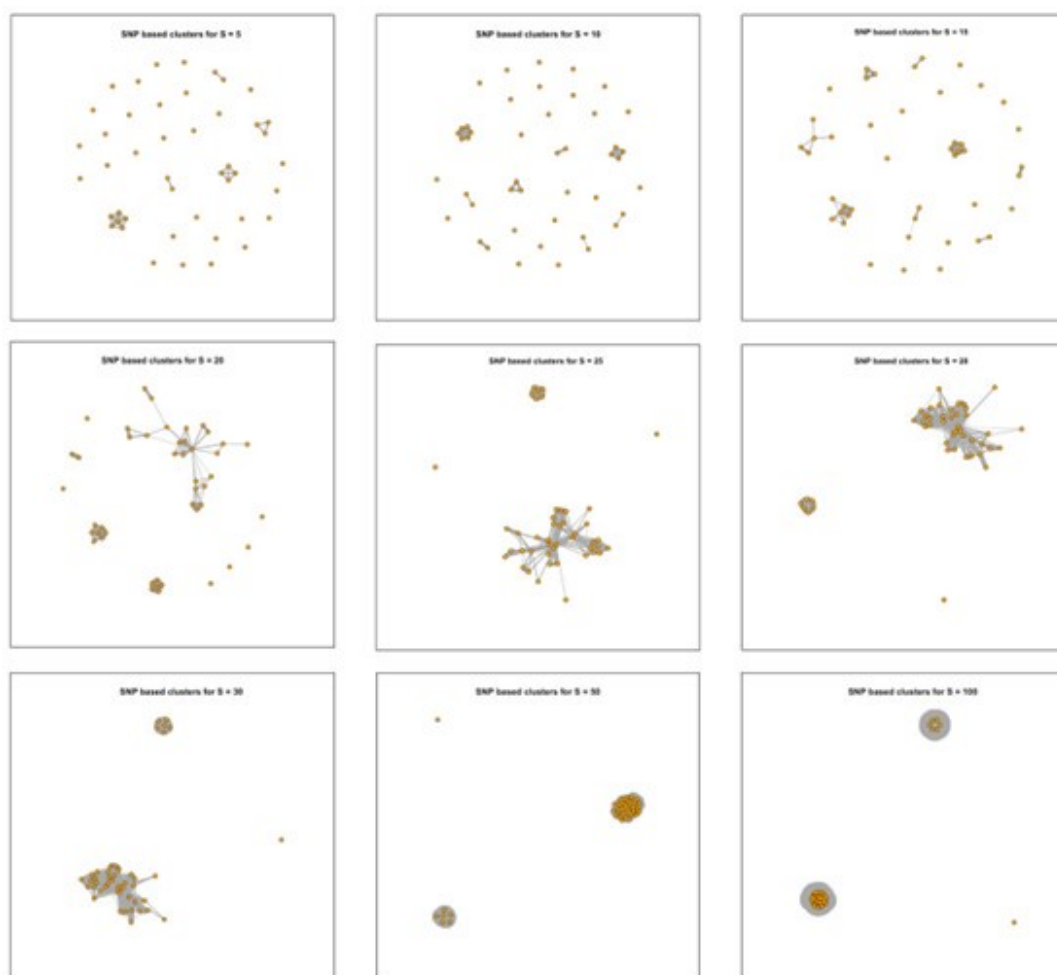
Using single nucleotide polymorphism (SNP) distance to trace pathogen transmission is a common approach (J Infect Dis. 2015 Apr 1;211(7):1154-63) and in our previous studies (hLife 2024; 2(5):246-256. mLif 2024; 3(1):156-160.). When the SNP distance within a cluster falls below a set threshold, the strains in that cluster are considered to have a potential direct transmission link. It is generally accepted that the lower the threshold, the more stringent the screening process becomes. However, there is little agreement in the literature regarding

what such a threshold should be, and the appropriate SNP cut-off for inferring transmission likely depends critically on the context (Mol Biol Evol. 2019 Mar 1;36(3):587-603).

In this study, we compared various thresholds (SNPs = 5, 10, 20, 25, 30, 35, 40, 50, 100) to ensure clustering in an appropriate manner. First, we summarized the tracing results under each threshold (Author response image 4), which demonstrated that, regardless of the threshold used, all strains associated with transmission events originated from the same location (New Figure 3a).

Author response image 4.

Clustering results of 45 newly isolated *S. Gallinarum* strains using different SNP thresholds of 5, 10, 15, 20, 25, 28, 30, 50, and 100 SNPs. The nine subplots represent the clustering results under each threshold. Each point corresponds to an individual strain, and lines connect strains with potential transmission relationships.



In response to your comments regarding the evolutionary rate, we estimated the overall evolutionary rate of the *S. Gallinarum* using BEAST. We applied the methodology described by Arthur W. Pightling et al. (Front Microbiol. 2022 Jun 16; 13:797997). The numbers of SNPs per year were determined by multiplying the evolutionary rates estimated with BEAST by the number of core SNP sites identified in the alignments. We hypothesize that a slower evolutionary rate in bacteria typically requires a lower SNP threshold when tracing transmission events using SNP distance analysis. Pightling et al.'s previous research found an average evolutionary rate of 1.97 SNPs per year (95% HPD, 0.48 to 4.61) across 22 different

Salmonella serotypes. Our updated BEAST estimation for the evolutionary rate of *S. Gallinarum* suggests it is approximately 0.74 SNPs per year (95% HPD, 0.42 to 1.06). Based on these findings, and our previous experience with similar studies (mBio. 2023 Oct 31;14(5):e0133323.), we set a threshold of 5 SNPs in the revised manuscript.

Then, we adopted the newly established SNP distance threshold ($n=5$) to update Figure 3a and New Supplementary Figure 8. The heatmap on the far right of New Figure 3a illustrates the SNP distances among 45 newly isolated *S. Gallinarum* strains from two locations in Zhejiang Province (Taishun and Yueqing). New Supplementary Figure 8 simulates potential transmission events between the bvSP strains isolated from Zhejiang Province ($n=95$) and those from China with available provincial information ($n=435$). These analyses collectively demonstrate the localized transmission pattern of bvSP within China. Our analysis using the newly established SNP threshold indicates that the 45 strains isolated from Taishun and Yueqing exhibit a highly localized transmission pattern, with pairs of strains exhibiting potential transmission events below the set threshold occurring exclusively within a single location. Subsequently, we conducted the SNP distance-based tracing analysis for the 95 strains from Zhejiang Province and those from China with available provincial information ($n=435$) (New Supplementary Figure 8, New Supplementary Table S8). Under the SNP distance threshold ($n=5$), we identified a total of 91 potential transmission events, all of which occurred exclusively within Zhejiang Province. No inter-provincial transmission events were detected. Based on these findings, we revised the methods and conclusions in the manuscript accordingly. We believe that the updated version well addresses your concerns.

Nevertheless, the final revised and updated results do not change the conclusions presented in our original manuscript. Instead, applying a more stringent SNP distance threshold allows us to provide solid evidence supporting the localized transmission pattern of *S. Gallinarum* in China.

(4) The HGT definition is non-standard, and phylogeny (vertical inheritance) is not controlled for.

The cited method:

'In this study, potentially recently transferred ARGs were defined as those with perfect identity (more than 99% nucleotide identity and 100% coverage) in distinct plasmids in distinct host bacteria using BLASTn (E-value {less than or equal to}10⁻⁵)'

This clearly does not apply here, as the application of distinct hosts and plasmids cannot be used. Subsequent analysis using this method is likely invalid, and some of it (e.g. Figure 6c) is statistically very poor.

Thank you for raising this important question. In our study, Horizontal Gene Transfer (HGT) is defined as the transfer of genetic information between different organisms, a process that facilitates the spread of antibiotic resistance genes (ARGs) among bacteria. This definition of HGT is consistent with that used in previous studies (Evol Med Public Health. 2015; 2015(1):193–194; ISME J. 2024 Jan 8;18(1):wrad032). In *Salmonella*, the transfer of antimicrobial resistance genes via HGT is not solely dependent on plasmids; other mobile genetic elements (MGEs), such as transposons, integrons, and prophages, also play significant roles. This has also been documented in our previous work (mSystems. 2023 Dec 21;8(6):e0088323). Given the involvement of various MGEs in the horizontal transfer of ARGs, we propose that the criteria for evaluating horizontal transfer via plasmids can also be applied to ARGs mediated by other MGEs.

In this study, we adopted stricter criteria than those used by Xiaolong Wang et al. Specifically, we defined two ARGs as identical only if they exhibited 100% nucleotide identity and 100% coverage. To address concerns regarding the potential influence of vertical inheritance in our

analysis, we have made the following improvements. In the revised manuscript, we provide a more detailed table that includes the co-localization analysis of each ARG with mobile genetic elements (New Supplementary Table 9). For prophages and plasmids, we required that ARGs be located directly within these elements. In contrast, for transposons and integrons, we considered ARGs to be associated if they were located within a 5 kb region upstream or downstream of these elements (Nucleic Acids Res. 2022 Jul 5;50(W1):W768-W773).

In the revised manuscript, we first categorized a total of 621 ARGs carried by 436 bvSP isolates collected in China according to the aforementioned criteria and found that 415 ARGs were located on MGEs. After excluding the ARGs not associated with MGEs, we recalculated the overall HGT frequency of 10 types of ARGs in China, the horizontal ARGs transfer frequency in three key regions, and the horizontal ARGs transfer frequency within a single region (New Supplementary Table 7). Based on the results, we updated relevant sections of the manuscript and remade Figure 6. The updated manuscript describes the results of this section as follows:

“Horizontal transfer of resistome occurs widely in localized bvSP

Horizontal transfer of the resistome facilitates the acquisition of AMR among bacteria, which may record the distinct acquisition event in the bacterial genome. To compare these events in a geographic manner, we further investigated the HGT frequency of each ARG carried by bvSP isolated from China and explored the HGT frequency of resistome between three defined regions. Potentially horizontally transferred ARGs were defined as those with perfect identity (100% identity and 100% coverage) and were located on MGEs across different strains (Fig. 6a). We first categorized a total of 621 ARGs carried by 436 bvSP isolates collected in China and found that 415 ARGs were located on MGEs. After excluding the ARGs not associated with MGEs, our findings reveal that horizontal gene transfer of ARGs is widespread among Chinese bvSP isolates, with an overall transfer rate of 92%. Specifically, 50% of the ARGs exhibited an HGT frequency of 100%, indicating that these ARGs might underwent extensive frequent horizontal transfer events (Fig. 6b). It is noteworthy that certain resistance genes, such as *tet(A)*, *aph(3'')-Ib*, and *aph(6)-Id*, appear to be less susceptible to horizontal transfer.

However, different regions generally exhibited a considerable difference in resistome HGT frequency. Overall, bvSP from the southern areas in China showed the highest HGT frequency (HGT frequency=95%). The HGT frequencies for bvSP within the eastern and northern regions of China are lower, at 92% and 91%, respectively (Fig. 6c). For specific ARG type, we found *tet(A)* is more prone to horizontal transfer in the southern region, and this proportion was considerably lower in the eastern region. Interestingly, certain ARGs such as *aph(6)-Id*, undergo horizontal transfer only within the eastern and northern regions of China (Fig. 6d). Notably, as a localized transmission pathogen, resistome carried by bvSP exhibited a dynamic potential among inter-regional and local demographic transmission, especially from northern region to southern region (HGT frequency=93%) (Fig. 6e, Supplementary Table 7).”

We also modified the current version of the pipeline used to calculate the HGT frequency of resistance genes. In the revised pipeline, users are required to provide a file specifying the locations of mobilome on the genome before formally calculating the HGT frequency of the target ARGs. The specific code and data used in the calculation have been uploaded to https://github.com/tjiaa/Cal_HGT_Frequency.

However, we also acknowledge that the current in silico method has some limitations. This approach heavily relies on prior information in existing resistome/mobilome databases. Additionally, the characteristics of second-generation sequencing data make it challenging to locate gene positions precisely. Using complete genome assemblies might be a crucial approach to address this issue effectively. In the revised manuscript, we have also provided a more detailed explanation of the implications of the current pipeline.

Regarding your second concern, "some of it (e.g., Figure 6c) is statistically very poor," the horizontal ARG transfer frequency calculation for each region was based on the proportion of horizontal transfer events of ARGs in that region to the total possible transfer events. As a result, we are unable to calculate the statistical significance between the two regions. Our aim with this approach is to provide a rough estimate of the extent of horizontal ARG transfer within the *S. Gallinarum* population in each region. In future studies, we will refine our conclusions by developing a broader range of evaluation methods to ensure more comprehensive assessment and validation.

(5) *Associations between lineages, resistome, mobilome, etc do not control for the effect of genetic background/phylogeny. So e.g. the claim 'the resistome also demonstrated a lineage-preferential distribution' is not well-supported.*

Thank you for your comments. We acknowledge that the associations between lineages and the mobilome/resistome may be influenced by the genetic background or phylogeny of the strains. For instance, our conclusion regarding the lineage-preferential distribution of the resistome was primarily based on New Figure 4a, where L3 is clearly shown to carry the most ARGs. Furthermore, we observed that L3b tends to harbor *_bla_{TEM-1B}*, *sul2*, and *tet(A)* more frequently than other lineages. However, we recognize that this evidence is insufficient to support a definitive conclusion of "demonstrated a lineage-preferential distribution". Therefore, we have re-examined the current manuscript and described these findings as a potential association between the mobilome/resistome and lineages.

(6) *The invasiveness index is not well described, and the difference in means is not biologically convincing as although it appears significant, it is very small.*

Thank you for pointing this out. For the invasiveness index mentioned in the manuscript, we used the method described in previous studies. (PLoS Genet. 2018 May 8;14(5), Nat Microbiol. 2021 Mar;6(3):327-338). Specifically, *Salmonella*'s ability to cause intestinal or extraintestinal infections in hosts is related to the degree of genome degradation. We evaluated the potential for extraintestinal infection by 45 newly isolated *S. Gallinarum* strains (L2b and L3b) using a model that quantitatively assesses genome degradation. We analyzed samples using the 196 top predictor genes, employing a machine-learning approach that utilizes a random forest classifier and delta-bitscore functional variant-calling. This method evaluated the invasiveness of *S. Gallinarum* towards the host, and the distribution of invasiveness index values for each region was statistically tested using unpaired *t*-test. The code used for calculating the invasiveness index is available at https://github.com/Gardner-BinfLab/invasive_salmonella. In the revised manuscript, we added a more detailed description of the invasiveness index calculation in the Methods section as follows:

Lines 592-603: "Specifically, *Salmonella*'s ability to cause intestinal or extraintestinal infections in hosts is related to the degree of genome degradation. We evaluated the potential for extraintestinal infection by 45 newly isolated *S. Gallinarum* strains (L2b and L3b) using a model that quantitatively assesses genome degradation. We analyzed each sample using the 196 top predictor genes for measuring the invasiveness of *S. Gallinarum*, employing a machine-learning approach that utilizes a random forest classifier and deltabitscore functional variant-calling. This method evaluated the invasiveness of *S. Gallinarum* towards the host, and the distribution of invasiveness index values for each region was statistically tested using unpaired *t*-test. The code used for calculating the invasiveness index is available at: https://github.com/Gardner-BinfLab/invasive_salmonella."

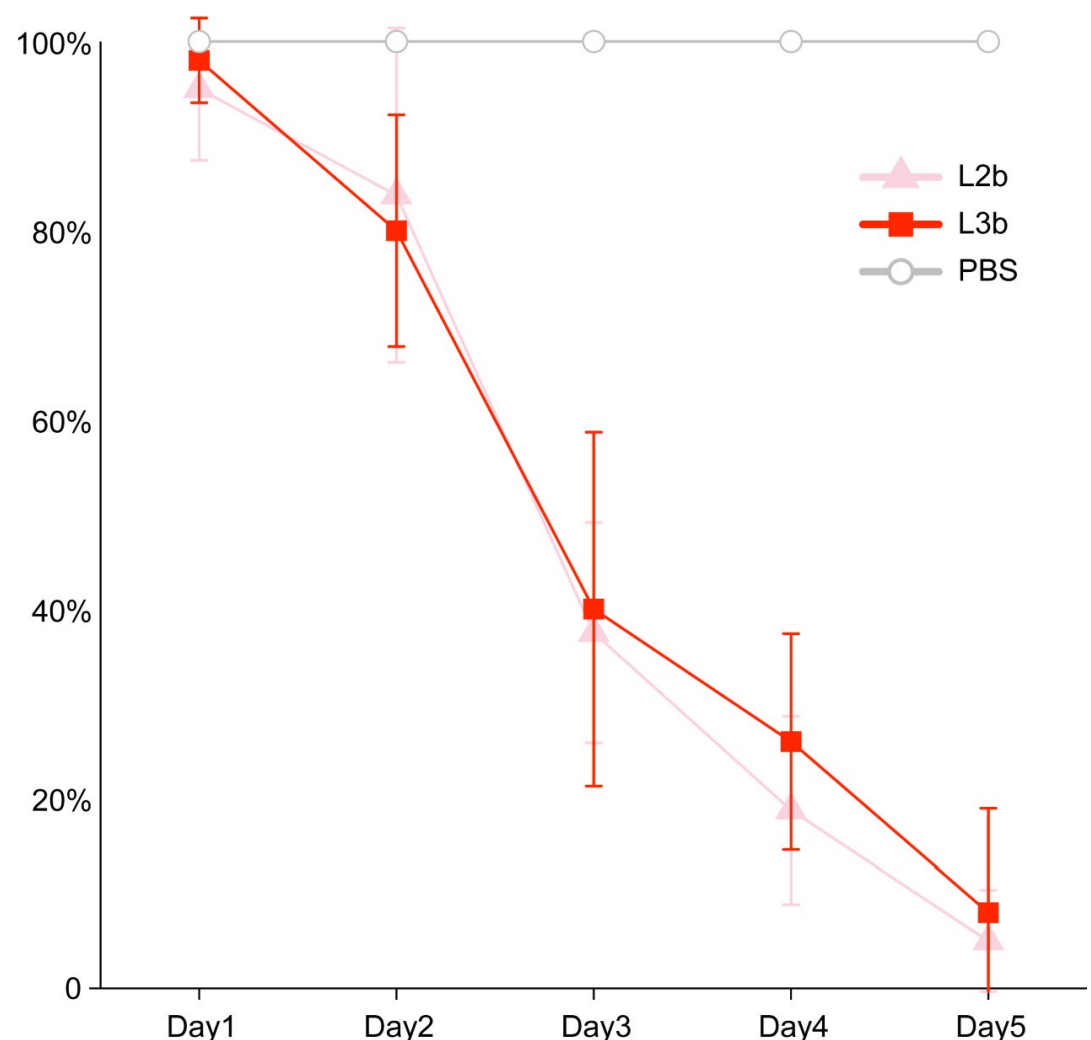
Regarding the second question, 'the difference in means is not biologically convincing as although it appears significant, it is very small,' we believe that this difference is biologically meaningful. In our previous work, we infected chicken embryos with different lineages of *S.*

Gallinarum (Natl Sci Rev. 2023 Sep 2;10(10):nwad228). The virulence of thirteen strains of *Salmonella* *Gallinarum*, comprising five from lineage L2b and eight from lineage L3b, was evaluated in 16-day-old SPF chicken embryos through inoculation into the allantoic cavity. Controls included embryos that inoculated with phosphate-buffered saline (PBS). The embryos were incubated in a thermostatic incubator maintained at 37.5°C with a relative humidity ranging from 50% to 60%. Prior to inoculation, the viability of the embryos was assessed by examining the integrity of their venous system and their movements; any dead embryos were excluded from the study. Overnight cultures resuspended in PBS at a concentration of 1000 CFU per 100 μ L were administered to the embryos. Mortality was recorded daily for a period of five days, concluding upon the hatching of the chicks.

It is generally accepted that strains with higher invasive capabilities are more likely to cause chicken embryo mortality. Our experimental results showed that the L2b, which exhibits higher invasiveness, with a slightly higher to cause chicken embryo death (Author response image 5).

Author response image 5.

The survival curves of chicken embryos infected with bvSP isolates from *S. Gallinarum* L2b and *S. Gallinarum* L3b. Inoculation with Phosphate Buffer Saline (PBS) were considered controls.



(7) *'In more detail, both the resistome and mobilome exhibited a steady decline until the 1980s, followed by a consistent increase from the 1980s to the 2010s. However, after the 2010s, a subsequent decrease was identified.'*

Where is the data/plot to support this? Is it a significant change? Is this due to sampling or phylogenetics?

Thank you for highlighting these critical points. The description in this statement is based on New Supplementary Figure 11. On the right side of New Supplementary Figure 11, we presented the average number of Antimicrobial Resistance Genes (ARGs) and Mobile Genetic Elements (MGEs) carried by *S. Gallinarum* isolates from different years, and we described the overall trend across these years. However, we realized that this statement might overinterpret the data. Given that this sentence does not impact our emphasis on the overall increasing trends observed in the resistome and mobilome, as well as their potential association, we decided to remove it in the revised manuscript.

The revised paragraph would read as follows:

Lines 261-268: "Variations in regional antimicrobial use may result in uneven pressure for selecting AMR. The mobilome is considered the primary reservoir for spreading resistome, and a consistent trend between the resistome and the mobilome has been observed across different lineages, from L1-L3c. We observed an overall gradual rise in the resistome quantity carried by bvSP across various lineages, correlating with the total mobilome content (S11 Fig). Furthermore, we investigated the interplay between particular mobile elements and resistome types in bvSP."

(8) *It is not clear what the burden of disease this pathogen causes in the population, or how significant it is to agricultural policy. The article claims to 'provide valuable insights for targeted policy interventions.', but no such interventions are described.*

Thank you for your constructive suggestions. *Salmonella Gallinarum* is an avian-specific pathogen that induces fowl typhoid, a severe systemic disease characterized by high mortality rates in chickens, thereby posing a significant threat to the poultry industry, particularly in developing countries (Rev Sci Tech. 2000 Aug;19(2):40524). In our previous research, we conducted a comprehensive meta-analysis of 201 publications encompassing over 900 million samples to investigate the global impact of *S. Gallinarum* (Sci Data. 2022 Aug 13;9(1):495). Our findings estimated that the global prevalence of *S. Gallinarum* is 8.54% (with a 95% confidence interval of 8.43% to 8.65%), with notable regional variations in incidence rates.

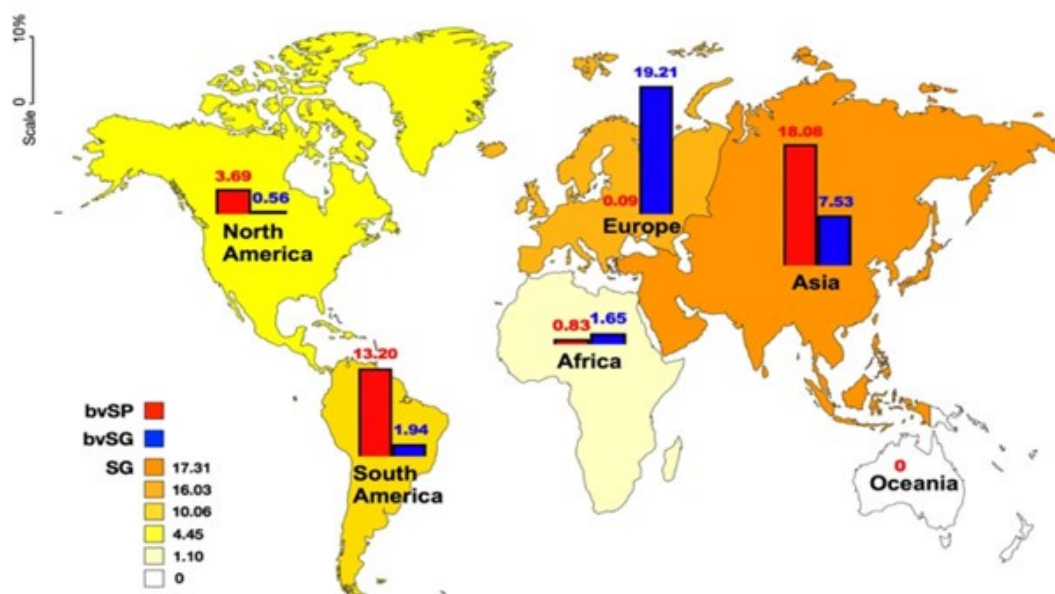
Our previously analysis focused on the prevalence of *S. Gallinarum* (including biovars SP and SG) across six continents. The results revealed that all continents, except Oceania, exhibited positive prevalences of *S. Gallinarum*. Asia had the highest prevalence at 17.31%, closely followed by Europe at 16.03%. In Asia, the prevalence of biovar SP was higher than that of biovar SG, whereas in Europe, biovar SG was observed to be approximately two hundred times more prevalent than biovar SP. In South America, the prevalence of *S. Gallinarum* was higher than that of biovar SP, at 10.06% and 13.20% respectively. Conversely, the prevalence of *S. Gallinarum* was relatively lower in North America (4.45%) compared to Africa (1.10%) (Author response image 6).

Given the significant economic losses caused by *S. Gallinarum* to the poultry industry and the potential risk of escalating antimicrobial resistance, more targeted policy interventions are urgently needed. Further elaboration on this implication is provided in the revised "Discussion" section as follows:

Lines 401-416: “In summary, the findings of this study highlight that *S. Gallinarum* remains a significant concern in developing countries, particularly in China. Compared to other regions, *S. Gallinarum* in China poses a notably higher risk of AMR, necessitating the development of additional therapies, i.e. vaccine, probiotics, bacteriophage therapy in response to the government's policy aimed at reducing antimicrobial use (J Infect Dev Ctries. 2014 Feb 13;8(2):129-36). Furthermore, given the dynamic nature of *S. Gallinarum* risks across different regions, it is crucial to prioritize continuous monitoring in key areas, particularly in China's southern regions where the extensive poultry farming is located. Lastly, from a One-Health perspective, controlling AMR in *S. Gallinarum* should not solely focus on local farming environments, with improved overall welfare on poultry and farming style. The breeding pyramid of industrialized poultry production should be targeted on the top, with enhanced and accurate detection techniques (mSphere. 2024 Jul 30;9(7):e0036224). More importantly, comprehensive efforts should be made to reduce antimicrobial usage overall and mitigate potential AMR transmission from environmental sources or other hosts (Vaccines (Basel). 2024 Sep 18;12(9):1067; Vaccines (Basel). 2023 Apr 18;11(4):865; Front Immunol. 2022 Aug 11;13:973224).”

Author response image 6.

A comparison of the global prevalence of *S. gallinarum* across continents.



(9) The abstract mentions stepwise evolution as a main aim, but no results refer to this.

Thank you for raising this issue. In the revised manuscript, we have changed “stepwise evolution” to simply “evolution” to ensure a more accurate and precise description.

(10) The authors attribute changes in population dynamics to normalisation in China-EU relations and hen fever. However, even if the date is correct, this is not a strongly supported causal claim, as many other reasons are also possible (for example other industrial processes which may have changed during this period).

Thank you for raising this critical issue. In the revised manuscript, we conducted a more stringent BEAST analysis for each lineage, as described earlier. This led to some changes in

the inferred evolutionary timelines. Consequently, we have removed the corresponding statement from the “Results” section. Instead, we now only provide a discussion of historical events, supported by literature, that could have facilitated the intercontinental spread of L2b and L3b in the “Discussion” section. We believe these revisions have made the manuscript more rigorous and precise.

Lines 332-342: “_The biovar types of *S. Gallinarum* have been well-defined as bvSP, bvSG, and bvSD historically (J Vet Med B Infect Dis Vet Public Health. 2005 Jun;52(5):2148). Among these, bvSP can be further subdivided into five lineages (L1, L2a, L2b, L3b, and L3c) using hierarchical Bayesian analysis. Different sublineages exhibited preferential geographic distribution, with L2b and L3b of bvSP being predominant global lineage types with a high risk of AMR. The historical geographical transmission was verified using a spatiotemporal Bayesian framework. The result shows that L3b was initially spread from China to Europe in the 18th-19th century, which may be associated with the European hen fever event in the mid-19th century (Burnham GP. 1855. The history of the hen fever: a humorous record). L2b, on the other hand, appears to have spread to Europe via South America, potentially contributing to the prevalence of bvSP in the United States.”

(11) No acknowledgment of potential undersampling outside of China is made, for example, 'Notably, all bvSP isolates from Asia were exclusively found in China, which can be manually divided into three distinct regions (southern, eastern, and northern).'

Perhaps we just haven't looked in other places?

We appreciate the reviewer's observation regarding the sampling distribution of isolates in this study. We acknowledge that while the isolates were collected from 15 different countries with, a significant proportion originated from China (Author response image 1). This focus is due to several reasons:

(1) As once a globally prevalent pathogen across the 20th century, *S. Gallinarum* was listed by the World Organization for Animal Health (WOAH) due to its economic importance. After 30 years of implementation the National Poultry Improvement Plan in the US, it was almost eradicated in high-income countries, and interestingly, it became an endemic pathogen with sporadic outbreaks in most low- or middle-income countries like China and Brazil. Given the vast expanse of China's land area and the country's economic factors, implementing the same measures remains a challenging endeavour.

(2) *S. Gallinarum* is an avian-specific pathogen, particularly affecting chickens, and its distribution is closely linked to chicken meat production in different countries. In some high chicken-producing developing countries, such as China and Brazil, there are more frequent reports of fowl typhoid. Data from the United States Department of Agriculture (USDA) on annual chicken meat production for 2023/2024 show that the global distribution of *S. Gallinarum* aligns closely with the overall chicken meat production of these countries (<https://fas.usda.gov/data/production/commodity/0115000>).

(3) Our primary objective was to investigate the localized resistome adaptation of *S. Gallinarum* in regions. Being a region with significant disease burden, China has reported numerous outbreaks (Sci Data. 2022 Aug 13;9(1):495; Sci Data. 2024 Feb 27;11(1):244) and a high AMR prevalence of this serovar (Natl Sci Rev. 2023 Sep 2;10(10):nwad228; mSystems. 2023 Dec 21;8(6):e0088323), making it an excellent example for understanding localized resistance mechanisms.

Nevertheless, a search of nearly a decade of literature on PubMed and a summary of the *S. Gallinarum* genome available on public databases indicate that the dataset used is the most complete. Furthermore, focusing on a specific region within China allowed us to conduct a detailed and thorough analysis. However, we highly agree that expanding the study to

include more isolates from other countries would enhance the generalizability of our findings, and we are actively collecting additional *S. Gallinarum* genome data. In the revised manuscript, we modified this sentence to indicate that this phenomenon is only observed in the current dataset, thereby avoiding an overly absolute statement:

Lines 131-135: “For the bvSP strains from Asia included in our dataset, we found that all originated from China. To further investigate the distribution of bvSP across different regions in China, we categorized them into three distinct regions: southern, eastern, and northern (Supplementary Table 3)”.

(12) *Many of the conclusions are highly speculative and not supported by the data.*

Thank you for your comment. We have carefully revised the manuscript to address your concerns. We hope that the changes made in the revised version meet your expectations and provide a clearer and more accurate interpretation of our findings.

(13) *The figures are not always the best presentation of the data:*

a. Stacked bar plots in Figure 1 are hard to interpret, the total numbers need to be shown.

Panel C conveys little information.

b. Figure 4B: stacked bars are hard to read and do not show totals.

c. Figure 5 has no obvious interpretation or significance.

Thank you for your comments. We have revised the figures to improve the clarity and presentation of the data.

In summary, the quality of analysis is poor and likely flawed (although there is not always enough information on methods present to confidently assess this or provide recommendations for how it might be improved). So, the stated conclusions are not supported.

Thank you for your valuable feedback. We have carefully revised the manuscript to address your concerns. We hope that the updated figures and tables, and new data in the revised version meet your expectations and provide more appropriate interpretation of our findings.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

This reviewer enjoyed reading this well-written manuscript. The authors are encouraged to address the following comments and revise the manuscript accordingly.

(1) Title: The authors use avian-restrict Salmonella to refer to Salmonella Gallinarum. Please consider using Salmonella Gallinarum in the title. Also, your analysis relates to resistome and mobilome. Would it make sense to add mobilome in the manuscript?

Thank you for your guidance. In the revised manuscript, we have changed the title to “Avian-specific *Salmonella enterica* Serovar *Gallinarum* transition to endemicity is accompanied by localized resistome and mobilome interaction”. We believe that this revised title more accurately reflects the content of our study.

(2) Abstract: This study uses 45 isolates from your labs. However, you failed to include these 45 isolates in the Abstract. Also, please clarify the sources of these isolates (from

dead chickens, or dead chicken embryos? You wrote in two different ways in this manuscript). Also, I am not entirely convinced how the results from these 45 isolates will support the overall conclusion of this work.

Thank you for your thorough review and constructive comments on the manuscript. In the revised version, we have added a description of 45 newly isolated *S. Gallinarum* strains in the Abstract to provide readers with a clearer understanding of the dataset used in this study.

Lines 36-41: “Using the most comprehensive whole-genome sequencing dataset of *Salmonella enterica* serovar Gallinarum (*S. Gallinarum*) collected from 16 countries, including 45 newly recovered samples from two related local regions, we established the relationship among avian-specific pathogen genetic profiles and localization patterns.”

Furthermore, the newly isolated *S. Gallinarum* strains were obtained from dead chicken embryos. We think your second concern may arise from the following description in the manuscript: “All 734 samples of dead chicken embryos were collected from Taishun and Yueqing in Zhejiang Province, China. After the thorough autopsy, the liver, intestines, and spleen were extracted and added separately into 2 mL centrifuge tubes containing 1 mL PBS. The organs were then homogenized by grinding.” In fact, all the collected dead chicken embryos were aged 19 to 20 days. At this developmental stage, collecting the liver, intestines, and spleen for isolation and cultivation of *S. Gallinarum* is possible. To avoid any confusion, we have included a more detailed description of the dead chicken embryos in the revised manuscript as follows:

Lines 447-451: “All 734 samples of dead chicken embryos aged 19 to 20 days were collected from Taishun and Yueqing in Zhejiang Province, China. After a thorough autopsy, the liver, intestines, and spleen were extracted and added separately into 2 mL centrifuge tubes containing 1 mL PBS. The organs were then homogenized by grinding.”

Regarding your concern about the statement, “I am not entirely convinced how the results from these 45 isolates will support the overall conclusion of this work,” we would like to clarify the significance of these new isolates. Our research first identified distinct characteristics in the 45 newly isolated *S. Gallinarum* strains from Taishun and Yueqing, Zhejiang Province. Specifically, we found that most of the strains from Yueqing belonged to sequence type ST92, whereas the majority from Taishun were ST3717. Additionally, there were significant differences between these geographically close strains in terms of SNP distance and predicted invasion capabilities. These findings suggest that *S. Gallinarum* may exhibit localized transmission patterns, which forms the basis of the scientific question and hypothesis we originally aimed to address. Furthermore, in our previous work, we collected 325 *S. Gallinarum* strains. By incorporating the newly isolated 45 strains, we aim to provide a more comprehensive view of the population diversity, transmission pattern and potential risk of *S. Gallinarum*. We will continue to endeavour to understand the global genomic and population diversity in this field.

Finally, we revised the sentences that could potentially raise concerns for readers:

Lines 175-177: “To investigate the dissemination pattern of bvSP in China, we obtained forty-five newly isolated bvSP from 734 samples (6.1% overall isolation rate) collected from diseased chickens at two farms in Yueqing and Taishun, Zhejiang Province.” > “To investigate the dissemination pattern of bvSP, we obtained forty-five newly isolated bvSP from 734 samples (6.1% overall isolation rate) collected from diseased chickens at two farms in Yueqing and Taishun, Zhejiang Province.”

(3) The manuscript uses nomenclature and classification into different sublineages. Did the authors establish the approaches for defining these sublineages in this group or did you follow the accepted standards?

Thank you very much for raising this important issue. The biovar types of *Salmonella* Gallinarum have historically been well-defined as *S. Gallinarum* biovar

Pullorum (bvSP), *S. Gallinarum* biovar Gallinarum (bvSG), and *S. Gallinarum* biovar Duisburg (bvSD) (J Vet Med B Infect Dis Vet Public Health. 2005 Jun;52(5):214-8). However, there seems to be no widespread consensus on the population nomenclature for the key biovar bvSP. In a previous study, Zhou et al. classified bvSP into six lineages:

L1, L2a, L2b, L3a, L3b, and L3c (Natl Sci Rev. 2023 Sep 2;10(10):nwad228). However, our more comprehensive analysis of *S. Gallinarum* using a larger dataset and hierarchical Bayesian clustering revealed that L3a, previously considered a distinct lineage, is actually a sublineage of L3c. Upon further review of our initial manuscript, we realized that the original submission did not strictly follow the lineage order proposed by Zhou et al. To avoid confusion in the typing system, we have adjusted the lineage nomenclature in the revised manuscript to reflect the corrected order (see Author response table 1).

(4) This reviewer is convinced with the analysis approaches and conclusion of this work.

*In the meantime, the authors are encouraged to discuss the application of the conclusion of this study: a) can the data be somehow used in the prediction model? b) would the conclusion from *S. Gallinarum* have generalized application values for other pathogens.*

Thank you for your constructive comments on the manuscript.

a) can the data be somehow used in the prediction model?

We believe that genomic data can be effectively used for constructing prediction models; however, the success of such models largely depends on the specific traits being predicted. In this study, we utilized a random forest prediction model based on 196 top genes (PLoS Genet. 2018 May 8;14(5)) to predict the invasiveness of 45 newly isolated strains. In relation to the antimicrobial resistance (AMR) issue discussed in this paper, we also conducted relevant analyses. For instance, we explored the use of image-based models to predict whether a genome is resistant to specific antibiotics (Comput Struct Biotechnol J. 2023 Dec 29;23:559-565). We are confident that the incorporation of newly generated data will facilitate the development of future predictive models, and we plan to pursue further research in this area.

*b) would the conclusion from *S. Gallinarum* have generalized application values for other pathogens.*

This might be explained from two perspectives. First, the key role of the mobilome in facilitating the spread of the resistome, as emphasized in this study, has also been confirmed in research on other pathogens (mBio. 2024 Oct 16;15(10):e0242824). Thus, we believe that the pipeline we developed to assess the horizontal transfer frequency of different resistance genes across regions applies to various pathogens. On the other hand, due to distinct evolutionary histories, different pathogens exhibit varying levels of adaptation to their environments. In this study, we found that *S. Gallinarum* tends to spread highly localized; however, this conclusion may not necessarily hold for other pathogens.

Reviewer #2 (Recommendations for the authors):

The authors would need to:

(1) Address my concerns about genomic analyses listed in the public review.

Thank you for your valuable feedback. We have carefully reviewed your concerns and made the necessary revisions to address the points raised about genomic analyses in the public review. We sincerely hope that these modifications meet your expectations and provide more robust analysis. We appreciate your thoughtful input and remain open to further suggestions to improve the manuscript.

| (2) *Add more detail on the genomic methods and their outputs, as suggested above.*

We have added further details to clarify the methodologies and outputs as mentioned above. Specifically, we expanded the description of the data processing, and the bioinformatic tools used for analysis. To ensure clarity, we also included an expanded discussion of the key outputs, highlighting their implications. We hope these revisions meet your expectations.

| (3) *Critically rewrite their introduction to make it clear what problem they are trying to address.*

Thank you for your guidance. In the revised manuscript, we have made the necessary modifications to the Introduction section to more clearly articulate the problem we aim to address.

| (4) *Critically rewrite their conclusions so they are supported by the data they present, and make it clear when claims are more speculative.*

Thank you for your guidance. In the revised manuscript, we have made the recommended modifications to the relevant sections of the conclusion as outlined above.

| *More minor issues I identified:*

| (1) *Typo in the title 'avian-restrict'.*

Done.

Line 1: “Avian-specific *Salmonella enterica* Seroovar Gallinarum transition to endemicity is accompanied by localized resistome and mobilome interaction.”

| (2) *'By utilizing the pipeline we developed' -- a pipeline has not been introduced at this point.*

In the revised manuscript, we have removed this section from the 'Abstract'.

Lines 46-48: “Notably, the mobilome-resistome combination among distinct lineages exhibits a geographical-specific manner, further supporting a localized endemic mobilome-driven process.”

| (3) *'has more than 90% serovars' -- doesn't make sense.*

Revised.

Lines 82-83: “*Salmonella*, a pathogen with distinct geographical characteristics, has more than 90% of its serovars frequently categorized as geo-serotypes.”

| (4) *'horrific mortality rates that remain a disproportionate burden'.*

Revised.

Lines 83-87: “Among the thousands of geo-serotypes, *Salmonella enterica* Sero var Gallinarum (*S. Gallinarum*) is an avian-specific pathogen that causes severe mortality, with particularly detrimental effects on the poultry industry in low- and middle-income countries.”

(5) *What is the rate, what is a comparison, how is it disproportionate?*

Thank you for your valuable feedback. It is challenging to accurately estimate the specific prevalence of *S. Gallinarum*, particularly due to the lack of comprehensive data in many countries. Numerous cases likely go unreported. However, *S. Gallinarum* is more commonly detected in low- and middle-income countries. Here, we provide three evidence supporting this observation. First, in our previous research, we conducted a comprehensive meta-analysis of 201 studies, involving over 900 million samples, to evaluate the global impact of *S. Gallinarum* (Sci Data. 2022 Aug 13;9(1):495). The estimated prevalence in 17 countries showed that Bangladesh had the highest rate (25.75%) of *S. Gallinarum* infections. However, for bio var Pullorum (bvSP), Argentina (20.69%) and China (18.18%) reported the highest prevalence rates. Second, previous studies have also reported that *S. Gallinarum* predominantly occurs in low- and middleincome countries (Vet Microbiol. 2019 Jan;228:165-172; BMC Microbiol. 2024 Oct 18;24(1):414). Finally, *S. Gallinarum* was once a globally prevalent pathogen in the 20th century. Following the implementation of eradication programs in most high-income countries, it was listed by the World Organization for Animal Health and subsequently became an endemic pathogen with sporadic outbreaks. However, similar eradication efforts are challenging to implement in low- and middle-income countries, leading to a disproportionately higher incidence of *S. Gallinarum* in these regions.

In the revised manuscript, we have rephrased this sentence to enhance its accuracy:

Lines 83-87: “Among the thousands of geo-serotypes, *Salmonella enterica* serovar Gallinarum (*S. Gallinarum*) is an avian-specific pathogen that causes severe mortality, with particularly detrimental effects on the poultry industry in low- and middle-income countries.”

(6) *'we collected the most comprehensive set of 580 S. Gallinarum isolates', -> 'we collected the most comprehensive set S. Gallinarum isolates, consisting of 580 genomes'.*

Revised.

Lines 97-100: “To fill the gaps in understanding the evolution of *S. Gallinarum* under regional-associated AMR pressures and its adaptation to endemicity, we collected the most comprehensive set *S. Gallinarum* isolates, consisting of 580 genomes, spanning the period from 1920 to 2023.”

(7) *Sequence reads are not available, and use a non-standard database. The eLife policy states: 'Sequence reads and assembly must be included for reference genomes, while novel short sequences, including epitopes, functional domains, genetic markers and haplotypes should be deposited, together with surrounding sequences, into Genbank, DNA Data Bank of Japan (DDBJ), or EMBL Nucleotide Sequence Database (ENA). DNA and RNA sequencing data should be deposited in NCBI Trace Archive or NCBI Sequence Read Archive (SRA).' So the sequences assemblies and reads should ideally be mirrored appropriately.*

Thank you for your valuable suggestion regarding submitting the genome data for the newly isolated 45 *S. Gallinarum* strains. The genome data have been deposited in the NCBI Sequence Read Archive (SRA) under two BioProjects. The “SRA Accession number” for each strain have been added to New Supplementary Table 1. We believe this will ensure that the data are more

readily accessible to a broader audience of researchers for download and analysis. We have revised the corresponding paragraph in the manuscript as follows:

Lines 606-608: “For the newly isolated 45 strains of *Salmonella* Gallinarum, genome data have been deposited in NCBI Sequence Read Archive (SRA) database. The “SRA Accession” for each strain are listed in Supplementary Table 1.”

| (8) *You should state at the start of the results which data is public, and how much is newly sequenced.*

Revised.

Lines 109-112: “To understand the global geographic distribution and genetic relationships of *S. Gallinarum*, we assembled the most comprehensive *S. Gallinarum* WGS dataset (n=580), comprising 535 publicly available genomes and 45 newly sequenced genomes.”

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